Ginzburg-Landau process and Glauber dynamics of critical Ising model on $\mathbb{Z}^d$

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Introduction

In this thesis, we study the convergence properties of two types of Markovian dynamics (or interacting particle systems) on $\mathbb{Z}^d$— the Kawasaki dynamics of ∞-temperature Ginzburg-Landau model in $d \geq 1$, and the Glauber dynamics of critical Ising model in high dimensions $d \geq 5$. Our main results show that both dynamics exhibit L^2-ergodicity and a polynomial rate of relaxation to equilibrium.

Theorem 0.0.1 ([LY03])**.** *Fix any $d \geq 1$, let $(\mathcal{P}_t)_{t\geq 0}$ denote the Markov semigroup for the Ginzburg-Landau process (1.3) in $\mathbb{Z}^d$ with invariant measures ν_ω (1.8). For any $F \in L^2(\nu_\omega)$ with $|||F||| < \infty$ (see (1.12) for definition of the triple norm), there exists a constant $C = C(d, \omega, V)$ such that*

$$\mathrm{Var}_{\nu_\omega}(\mathcal{P}_t F) \leq \frac{C\,|||\,F\,|||^2 \log(2+t)}{(1+t)^{d/2}}. \tag{0.1}$$

In particular, $\mathrm{Var}_{\nu_\omega}(F) := \mathbb{E}_{\nu_\omega}(F - \mathbb{E}_{\nu_\omega} F)^2$ denotes the variance for any $L^2(\nu_\omega)$ test function F.

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Theorem 0.0.2 ([Hu25]). *Fix $d \geq 5$, let $(P_t)_{t\geq 0}$ denote the infinite-volume Markov semigroup for the Ising Glauber dynamics at critical temperature β_c. Given (3.44), and conditional on Assumption 3.3.1, there exists a constant $C = C(\beta_c, \eta, \delta, d, c_M)$ such that the polynomial bound*

$$\langle \sigma_0, P_t \sigma_0 \rangle_{\mathbb{Z}^d} \leq \frac{C}{t^\alpha}, \quad \forall t \geq 0 \tag{0.2}$$

holds with the exponent α given by

$$\alpha = \frac{2\delta \wedge 1}{2\eta}. \tag{0.3}$$

In the above, $c_M > 0$ is an upper bound for the flipping rates of the dynamics, see (1.17). $\langle \cdot \rangle_{\mathbb{Z}^d}$ denotes the unique infinite volume Ising expectation (1.14) on $\mathbb{Z}^d$ at critical temperature β_c. $\langle f, g \rangle_{\mathbb{Z}^d} := \langle fg \rangle_{\mathbb{Z}^d}$ is the L^2 inner product for any mean zero functions f and g.

Notations

We denote general finite subgraphs of the integer lattice by $\Lambda \subset \mathbb{Z}^d$. In results where the geometry of the underlying graph is relevant, we focus on two specific cases: (1) The finite cube $\Gamma_L = \{1, ..., L\}^d$ of side-length L; (2) Hypercubes centered at the origin, denoted by $\Lambda_L := \{-L, ..., L\}^d$, along with their translations: $x + \Lambda_L := \{x - L, ..., x + L\}^d$, $\forall x \in \mathbb{Z}^d$. Two vertices $x, y \in \mathbb{Z}^d$ are called adjacent iff $\sum_{i=1}^d |x_i - y_i| = 1$ and this is denoted by $x \sim y$. For the thermodynamic limits of such systems, we refer to both the infinite-volume limit: $\lim_{L\to\infty} \Lambda_L = \mathbb{Z}^d$, and the asymptotic statistics of the spins. An unbounded continuous spin field on Λ is represented by a function $\varphi : \Lambda \to \mathbb{R}$, while a discrete Ising spin configuration is given by $\sigma : \Lambda \to \{\pm 1\}$. We will also use Greek letters ν and μ to denote the Gibbs measures for φ and σ, respectively. A classical observable $F : \mathbb{R}^\Lambda \to \mathbb{R}$ (or $F : \{\pm 1\}^\Lambda \to \mathbb{R}$) is said to be a local function if the values of F depend only on a finite number of spins $\{\varphi_1, ..., \varphi_N\}$ (or $\{\sigma_1, ..., \sigma_N\}$).

For simplicity of notation, we sometimes write $X \lesssim Y$ as shorthand for inequality $X \leq cY$ for some constant $c > 0$. If X/Y is bounded away from 0 and ∞, that is, if there exist finite constants c, C such that $0 < c \leq X/Y \leq C < \infty$, we write $X \asymp Y$. Furthermore, we write $\lesssim_a$ and $\asymp_a$ if the constants c and C in the asymptotic relations are functions of a certain parameter a. Last, we use the letters $\mathcal{L}$, $\mathcal{P}$, and $\mathcal{D}$ to denote the generators, semigroups, and Dirichlet forms for the Ginzburg-Landau process, while for the Ising dynamics we write L, P, and D, respectively.

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1 Preliminaries on Gibbs measures and stochastic dynamics

The Gibbs measure of a (classical) lattice statistical mechanics model is a probability distribution on the set of all spin fields (or discrete spin configurations). This distribution describes the probability of observing a

particular microscopic field φ (or configuration σ) in the equilibrium state. Given any test function F, the expectation of such distributions is formally given by

$$\mathbb{E}_\nu[F(\varphi)] = \frac{1}{Z}\int_{\mathbb{R}^\Lambda} e^{-H(\varphi)}F(\varphi)d\varphi, \quad \text{or} \quad \mathbb{E}_\mu[F(\sigma)] = \frac{1}{Z}\sum_{\sigma\in\{\pm1\}^\Lambda} e^{-H(\sigma)}F(\sigma) \tag{1.1}$$

for continuous and discrete spins, respectively. In particular, the normalizing constant Z refers to the partition function and H refers to the Hamiltonian (or action) of the system which describes the coupling behavior of the individual spins.

Motivated by the theory of integrable models, solid-state physics, interacting quantum field theory, log-gases and random matrices [Bax82, DMS97, GJ87, Sim74, Sim93, For10], we consider a family of Hamiltonian functionals H of the form:

$$H(\varphi) = \frac{\beta}{2}\sum_{\substack{x\sim y\\ x,y\in\Lambda}}(\varphi_x-\varphi_y)^2 + V(\varphi). \tag{1.2}$$

The first term in the expression above describes a discrete Gaussian free field (DGFF) [FV18, WP21], with inverse temperature $\beta = 1/T$ that causes the phase transition phenomena. The second term, $V(\varphi)$, describes the interactions of certain quantum field theory models. For example, choosing $V(\varphi) = \sum_{x\in\Lambda}[g\varphi(x)^4 + a\varphi(x)^2]$ and $V(\varphi) = \sum_{x\in\Lambda}\cos[\sqrt{\lambda}\varphi(x)]$ corresponds to the φ^4 model and the sine-Gordon model, respectively; see [BBS19].

The Hamiltonian (1.2) has deep connections to the models of interest in this paper. First, the Ising model (1.13) can be viewed as a σ-model approximation of the lattice φ^4 theory [Spe12]. On the other hand, if the potential functional V is chosen to be a strictly convex function of the field variables, the corresponding Hamiltonian is said to be the Ginzburg-Landau $\nabla\varphi$ model. The Ginzburg-Landau effective theory was first introduced by Zamolodchikov to study a special class of minimal models (see [DMS97] for an overview). Within the framework of statistical mechanics and probability theory, the Ginzburg-Landau model is also known as the anharmonic crystal [Fun02, Mil11], since the field variable φ represents the heights of a random surface that serves as an interface separating two distinct phases. For the study of stochastic dynamics, we simplify the Ginzburg-Landau model to be a product measure on $\mathbb{Z}^d$ (1.8). This can be interpreted as taking the ∞-temperature limit (i.e., $\beta = 0$) in (1.2).

1.1 Ginzburg-Landau process

Consider continuous spin fields $\varphi \in \mathbb{R}^{\mathbb{Z}^d}$ defined in the full d-dimensional integer lattice $\mathbb{Z}^d$. For each site $x \in \mathbb{Z}^d$, we construct an additive perturbation of the quadratic potential $V(x) = \frac{1}{2}x^2 + U(x)$, where $U : \mathbb{R} \to \mathbb{R}$ is a smooth bounded function such that $\|U'\|_\infty < \infty$, $\int_{\mathbb{R}}\exp[-U(x)]dx = 1$. The infinite-volume Ginzburg-Landau process is defined as a diffusion process $\varphi(t)$ given by the SDE:

$$d\varphi_t(x) = \sum_{i=1}^d \big(V'[\varphi_{x+e_i}(t))] + V'[\varphi_{x-e_i}(t))] - 2V'[\varphi_x(t))]\big)dt + \sum_{i=1}^d [dB_t(x+e_i) - dB_t(x)]. \tag{1.3}$$

In particular, $e_i, 1 \le i \le d$ refers to the unit vector in the i-th direction, and $B_t(x), x \in \mathbb{Z}^d$ are independent Brownian motions. Given any local function $g : \mathbb{R}^{\mathbb{Z}^d} \to \mathbb{R}$, one may calculate the generator of (1.3) as:

$$(\mathcal{L}g)(\varphi) = \frac{1}{2}\sum_{x\sim y}(\partial_x-\partial_y)^2 g(\varphi) - \frac{1}{2}\sum_{x\sim y}[V'(\varphi_y)-V'(\varphi_x)](\partial_y-\partial_x)g(\varphi). \tag{1.4}$$

The infinite-volume generator can be further restricted to the finite cube $\Gamma_L := \{1,...,L\}^d$:

$$(\mathcal{L}_{\Gamma_L}g)(\varphi) = \frac{1}{2}\sum_{\substack{x\sim y\\ x,y\in\Gamma_L}}(\partial_x-\partial_y)^2 g(\varphi) - \frac{1}{2}\sum_{\substack{x\sim y\\ x,y\in\Gamma_L}}[V'(\varphi_y)-V'(\varphi_x)](\partial_y-\partial_x)g(\varphi). \tag{1.5}$$

The expressions for (1.4) and (1.5) suggest that the infinitesimal motion of the dynamics is essentially the interchanging of particles and potential energy for all pairs of nearest-neighbor vertices. Thus, such dynamics are said to be conservative.

Motivated by the study of hydrodynamic large deviations (see, for example, [Var93]) of such interacting particle systems, we consider the tilting of the potential by any real parameter $\lambda \in \mathbb{R}$. More precisely, denote by $Z : \mathbb{R} \to \mathbb{R}$ the partition function

$$Z(\lambda) = \int_{\mathbb{R}} e^{\lambda x - V(x)} dx. \tag{1.6}$$

Let $R(\lambda) = \partial_\lambda \log Z(\lambda)$, which is a smooth and strictly increasing function with inverse Υ. For any $\omega \in \mathbb{R}$, it is easy to see that

$$\omega = \frac{1}{Z(\Upsilon(\omega))} \int_{\mathbb{R}} x e^{\Upsilon(\omega)x - V(x)} dx. \tag{1.7}$$

The infinite system of SDEs (1.3) admit a family of invariant product measures indexed by λ. Let $H_\lambda(\varphi) = \prod_{x\in\mathbb{Z}^d}[V(\varphi_x) - \lambda\varphi_x]$ be the Hamiltonian, we denote by $\bar{\nu}_\lambda$ the infinite-volume grand canonical measure for the Ginzburg-Landau model

$$\bar{\nu}_\lambda(d\varphi) := \frac{1}{Z(\lambda)} e^{-H_\lambda(\varphi)} d\varphi = \bigotimes_{x\in\mathbb{R}^d} \frac{1}{Z(\lambda)} e^{\lambda\varphi_x - V(\varphi_x)} d\varphi_x. \tag{1.8}$$

If we define $\nu_\omega := \bar{\nu}_\lambda$, then $\mathbb{E}_{\nu_\omega}(\varphi_x) = \omega$ for all $\omega \in \mathbb{R}$ and $x \in \mathbb{Z}^d$. Moreover, we see that the product measures $\bar{\nu}_\lambda$ (or ν_ω) are reversible for the Markovian dynamics (1.3).

In Section 2.1, we will prove the log-Sobolev inequality (2.1) for the Ginzburg-Landau process in $\Gamma_L = \{1, ..., L\}^d$. Since (1.8) are product measures, we restrict them to Γ_L and denote the truncations by $\bar{\nu}_\lambda^{\Gamma_L}$ (or $\nu_\omega^{\Gamma_L}$), in analogy with the restriction of the generator (1.5). From the perspectives of statistical physics, the total number of particles $\sum_{x\in\Gamma_L} \varphi_x$ is arbitrary for $\bar{\nu}_\lambda^{\Gamma_L}$ or $\nu_\omega^{\Gamma_L}$, hence they are also known as the grand canonical measures. Notice that the Kawasaki dynamics (1.3) preserves hyperplanes in $\mathbb{R}^{\Gamma_L}$ with a fixed number of particles, we introduce the canonical measure $\nu_M^{\Gamma_L}$ for each $M \in \mathbb{R}$:

$$\nu_M^{\Gamma_L}(\cdot) := \nu_\omega^{\Gamma_L}\Big(\cdot \,\Big|\, \sum_{x\in\Gamma_L} \varphi_x = M\Big). \tag{1.9}$$

The Kawasaki Dirichlet form $\mathcal{D}_\nu^L$, associated with $\mathcal{L}_{\Gamma_L}$, and any measure $\nu = \nu_\omega^{\Gamma_L}$, ν_ω, or $\nu_M^{\Gamma_L}$, is defined by

$$\mathcal{D}_\nu^L(F) := -\mathbb{E}_\nu[F \cdot (\mathcal{L}_{\Gamma_L} F)] = \frac{1}{2} \sum_{\substack{x\sim y \\ x,y\in\Gamma_L}} \mathbb{E}_\nu[(T^{x,y}F)^2], \tag{1.10}$$

where $F \in L^2(\nu)$, $T^{x,y}F = \partial_x f - \partial_y f$, and the second identity follows easily from integration by parts. Due to independence, measures $\nu_\omega^{\Gamma_L}$ and ν_ω give the same expectation and Dirichlet form on Γ_L. Moreover, we may also define Dirichlet forms for more general unbounded essentially self-adjoint operators $\mathcal{L}$ to be

$$\mathcal{D}_\nu(F) := -\mathbb{E}_\nu[F \cdot (\mathcal{L}F)] \tag{1.11}$$

Without causing any confusion, we still denote them as $\mathbb{E}_{\nu_\omega}(\cdot)$ and $\mathcal{D}_{\nu_\omega}^L(\cdot)$. For local observables F, the function $\mathcal{L}F$ is always well defined. To extend the domain of the infinitesimal generator $\mathcal{L}$, we consider the set of functions $F \in \mathbb{R}^{\mathbb{Z}^d}$ such that

$$|||F||| := \sum_{x\in\mathbb{Z}^d} \|(\partial_x f)(\varphi)\|_\infty < \infty. \tag{1.12}$$

Heuristically speaking, one should interpret functions with finite triple norm to be vanishing at infinity.

1.2 Critical Ising model and its Glauber dynamics

Let Λ be a finite subset of $\mathbb{Z}^d, d \geq 2$. At fixed inverse temperature $\beta = 1/T$, τ-boundary condition, and zero external field, we define the finite volume Ising model on spin configurations $\sigma \in \Omega_\Lambda = \{-1, +1\}^\Lambda$ to be the probability measure

$$\mu^\tau_{\Lambda,\beta}(\sigma) = \frac{1}{Z^\tau_{\Lambda,\beta}} \exp\Big(\beta \sum_{\substack{x\sim y \\ x,y\in\Lambda}} \sigma_x\sigma_y + \beta \sum_{\substack{a\sim b \\ a\in\Lambda,\ b\in\partial\Lambda}} \sigma_a\tau_b\Big). \tag{1.13}$$

Here $\partial\Lambda = \{x \in \mathbb{Z}^d; x \sim \Lambda, x \notin \Lambda\}$, $\tau \in \Omega_{\mathbb{Z}^d} = \{-1, +1\}^{\mathbb{Z}^d}$ or $\tau \equiv 0$, $Z^\tau_{\Lambda,\beta}$ denotes the partition function. The Ising model is said to be under $+$ boundary, $-$ boundary, and free boundary condition if we take $\tau \equiv 1$, $\tau \equiv -1$ and $\tau \equiv 0$ in (1.13) respectively. For the free boundary case, we will omit the superscripts τ.

It can be proved that, for Ising models in dimensions $d \geq 2$, the thermodynamic limit of the Gibbs measure (1.13) is unique for all $\beta \in [0, \beta_c)$, and not unique for $\beta \in (\beta_c, \infty)$. This is called the first-order phase transition of the Ising model [FV18, Definition 3.24], and the threshold $\beta_c(d) > 0$ is known as the critical inverse temperature. Since we will be mostly considering the critical phase in this article (except for the proof of Theorem 2.2.1), the beta subscript will also be omitted in the following context. The expectation under this finite volume measure is mostly denoted by $\mathbb{E}^\tau_\Lambda(\cdot)$ or $\langle\cdot\rangle^\tau_\Lambda$, and the variance is denoted by $\mathrm{Var}^\tau_\Lambda(\cdot)$. Due to the continuity of phase transition (see [ADCS15] for $d \geq 3$, and Kaufman-Onsager's approach for $d = 2$), for any sequence of finite regular subgraphs $\Lambda_n \uparrow \mathbb{Z}^d$ and boundary conditions $(\tau_n)_{n\geq 1}$, the distributions $\langle\cdot\rangle^{\tau_n}_{\Lambda_n}$ weakly converge to a unique infinite-volume Gibbs state $\langle\cdot\rangle_{\mathbb{Z}^d}$. More precisely, for any local function $F : \Omega_{\mathbb{Z}^d} \mapsto \mathbb{R}$, we have

$$\lim_{n\to\infty} \langle F\rangle^{\tau_n}_{\Lambda_n} = \langle F\rangle_{\mathbb{Z}^d}. \tag{1.14}$$

The Gibbs state $\langle\cdot\rangle_{\mathbb{Z}^d}$ implicitly characterizes the unique infinite volume Ising measure on the space $\Omega_{\mathbb{Z}^d}$. For more details, see [FV18].

The Glauber dynamics of critical Ising model is a family of Markov processes determined by infinitesimal generators on certain function spaces. First, for the finite volume case, we consider the space of square-integrable functions $L^2(\mu^\tau_\Lambda)$. Define the self-adjoint generator L^τ_Λ of the finite volume Glauber dynamics by

$$(\mathrm{L}^\tau_\Lambda F)(\sigma) := \sum_{x\in\Lambda} c(x,\sigma)(\nabla_x F)(\sigma) = \sum_{x\in\Lambda} c(x,\sigma)[F(\sigma^x) - F(\sigma)], \quad \forall F \in L^2(\mu^\tau_\Lambda). \tag{1.15}$$

where σ^x denotes the spin configuration obtained by flipping σ at vertex $x \in \Lambda$. The coefficients $c(x,\sigma)$ are called the flipping rates, and are additionally required to satisfy the following:

(i) Finite range interactions: For some fixed $r > 0$ and any $x \in \Lambda$, if the spin configurations $\sigma, \sigma' \in \Omega_\Lambda$ agree on $\{y \in \Lambda; \max_{1\leq i\leq d} |y_i - x_i| \leq r\}$, then $c(x,\sigma) = c(x,\sigma')$.

(ii) Detailed balance: For all $\sigma \in \Omega_\Lambda$ and $x \in \Lambda$,

$$\frac{c(x,\sigma)}{c(x,\sigma^x)} = \exp\big(2\beta_c\sigma_x \sum_{y\sim x} \sigma_y\big). \tag{1.16}$$

(iii) Positivity and boundedness: There exists positive constants c_m and c_M such that

$$0 < c_m \leq \inf_{x,\sigma} c(x,\sigma) \leq \sup_{x,\sigma} c(x,\sigma) \leq c_M < \infty. \tag{1.17}$$

(iv) Translation invariance: If there exist some $k \in \Lambda$ such that $\sigma_y = \sigma'_{y+k}$ for all $y \in \Lambda$, then $c(x,\sigma) = c(x+k,\sigma')$.

Two most common examples for the choice of flipping rates are

(i) Metropolis: $c(x,\sigma) = \exp\left(2\beta_c \sigma_x \sum_{y\sim x} \sigma_y\right) \wedge 1$.

(ii) Heat-bath: $c(x,\sigma) = \left[1 + \exp\left(-2\beta_c \sum_{y\sim x} \sigma_y\right)\right]^{-1}$.

One could naturally extend this finite-volume Glauber dynamics to the infinite volume case. Similar to the continuous case, let $B(\Omega_{\mathbb{Z}^d})$ be a function space consisting of the family of functions $F : \Omega_{\mathbb{Z}^d} \mapsto \mathbb{R}$ such that $|||F||| < \infty$, where the triple norm is given by

$$|||F||| := \sum_{x\in\mathbb{Z}^d} \sup_{\sigma\in\Omega_{\mathbb{Z}^d}} |(\nabla_x f)(\sigma)| < \infty. \tag{1.18}$$

We then naturally define the infinite-volume Glauber generator by

$$(\mathrm{L}F)(\sigma) = \sum_{x\in\mathbb{Z}^d} c(x,\sigma)(\nabla_x F)(\sigma), \quad \forall F \in B(\Omega_{\mathbb{Z}^d}). \tag{1.19}$$

In this case, L is an essentially self-adjoint operator on $L^2(\mu_{\mathbb{Z}^d})$ and the dynamics is reversible with respect the infinite-volume Gibbs state $\langle\cdot\rangle_{\mathbb{Z}^d}$.

For the finite-volume dynamics, the Glauber Dirichlet form associated with the generator (1.15) is

$$\mathrm{D}_\Lambda^\tau(F) = -\mathbb{E}_\Lambda^\omega[F \cdot \mathrm{L}_\Lambda^\tau F]. \tag{1.20}$$

The infinite-volume Dirichlet form is defined similarly by replacing the generator by L.

1.3 Functional inequalities

For the thermodynamical evolution of finite-volume statistical mechanics models (such as the aforementioned Ginzburg-Landau processes and Glauber Ising dynamics), a very natural question to investigate is the quantitative uniform ergodicity: $\mathcal{P}_t F \to \mathbb{E}_\nu(F)$. To be more specific, for any equilibrium spin field distribution ν (or discrete spin measure μ) on a finite subgraph Λ of the integer lattice, let $\{\mathcal{P}_t := e^{t\mathcal{L}}, t \geq 0\}$ denote the Markov semigroup associated with generator $\mathcal{L}$, and $F \in L^2(\nu)$ (or $F \in L^2(\mu)$ in the Ising case). Can we find a rate function $\gamma(t)$ such that $\lim_{t\to\infty} \gamma(t) = 0$ and

$$\|\mathcal{P}_t F - \mathbb{E}_\nu(F)\|_\infty \leq \gamma(t)\, |||\, F|||?$$

In finite volumes, the validity of the classical Sobolev inequality (which is equivalent to ultracontractivity, see [GZ03]) enables one to first bound the uniform norms $\|F\|_\infty$ by the $L^2(\nu)$ norms $\|F\|_2$. As a result, we focus on the $L^2(\nu)$ ergodicity and consider the relative entropy (in this case, equivalent to the KL-Divergence):

$$H(\nu_t|\nu) = \mathbb{E}_\nu(F_t \log F_t), \tag{1.21}$$

where ν_t refers to the distribution of the Markov process at time t and $F_t := \frac{d\nu_t}{d\nu}$ is the Radon-Nikodym density. In general, if the non-negative function F is not normalized, i.e., $E_\nu(F) \neq 1$, we define the relative entropy to be

$$\mathrm{Ent}_\nu(F) = \mathbb{E}_\nu[\Phi(F)] - \Phi[E_\nu(F)], \quad \Phi(x) = x\log x. \tag{1.22}$$

By straightforward calculations, we obtain the following bound, which is very useful for the proof of our main results.

Lemma 1.3.1 (de Bruijn inequality)**.** *For any N-dimensional ($N \geq 3$) stochastic dynamics with invariant distribution ν and Dirichlet form $\mathcal{D}_\nu(F) := -\mathbb{E}_\nu[F \cdot (\mathcal{L}F)]$, we have*

$$\partial_t \,\mathrm{Ent}_\nu(\mathcal{P}_t F) \leq -\mathcal{D}_\nu(\sqrt{\mathcal{P}_t F}). \tag{1.23}$$

We claim that the de Bruijn inequality, as well as the classical Sobolev inequality, together imply the $L^2(\nu)$ ergodicity. First, by Jensen's inequality and the concavity of $\log(x)$, we have

$$\mathbb{E}_\nu[\mathcal{P}_t F \cdot \log \mathcal{P}_t F] \lesssim_{p,N} \log \mathbb{E}_\nu[(\mathcal{P}_t F)^p] \lesssim_{p,N} a\mathcal{D}_\nu(\mathcal{P}_t F) + b\mathbb{E}_\nu[(\mathcal{P}_t F)^2], \tag{1.24}$$

where $a, b > 0$ are suitable constants, and $p = \frac{2N}{N-2}$. Next, (1.23), (1.24), and Gronwall's inequality yield

$$H(\nu_t|\nu) \le e^{-2ct} H(\nu_0|\nu). \tag{1.25}$$

Indeed, we have just seen that the classical Sobolev inequality implies the log-Sobolev inequality.

Definition 1.3.2 (log-Sobolev inequality)**.** The probability measure ν on $\mathbb{R}^\Lambda$ (or μ on $\{\pm 1\}^\Lambda$) is said to satisfy the log-Sobolev inequality (LSI), with respect to the Dirichlet form $\mathcal{D}_\nu(\cdot)$ (or $\mathrm{D}_\mu(\cdot)$) and constant $\gamma > 0$, if and only if for any smooth, non-negative local function F on $\mathbb{R}^\Lambda$ ($\{\pm 1\}^\Lambda$), we have

$$\mathrm{Ent}_\nu(F) \le \frac{2}{\gamma} \mathcal{D}_\nu(\sqrt{F}). \tag{LSI}$$

In particular, the largest constant $\gamma > 0$ for the above inequality to hold is called the log-Sobolev constant (LSI constant).

The log-Sobolev inequality further implies a broader class of functional inequalities. For example, given (LSI), the LSI constant $\gamma > 0$ is also valid as the spectral gap of the generator $\mathcal{L}$.

Corollary 1.3.3 (Spectral gap inequality)**.** *The log-Sobolev inequality with constant γ implies the spectral gap inequality (SGI):*

$$\mathrm{Var}_\nu(F) \le \frac{1}{\gamma} \mathcal{D}_\nu(F), \tag{SGI}$$

where $\mathrm{Var}_\nu(F)$ denotes the variance of F with respect to ν.

Remark 1.3.4. The log-Sobolev inequality is shown to be equivalent to the hypercontractivity property, as first proved in the seminal work [Gro76], also see [GZ03] for a detailed exposition. In fact, hypercontractivity was introduced first in the context of constructive quantum field theory, where this property leads to the asymptotic independence of free field operators, and henceforth the spatial cutoff for the interacting Hamiltonian functionals, see [GJ87] and [Sim74].

In the following, we discuss two important approaches to obtain finite-volume LSIs. The first sufficient criterion is known as the Bakry-Emery theorem [BE85]. The idea is non-perturbative, but it only holds when the spin measure is strictly log-concave and the single spin space admits a Riemannian manifold structure. The proof of Bakry-Emery's criterion is based on the Γ_2-calculus, see, for example [Led01-1].

Theorem 1.3.5 (Bakry-Emery)**.** *Let ν be a probability measure on $\mathbb{R}^N$ defined as the first formula in (1.1), and assume that there exists $\lambda > 0$ such that Hessian of the Hamiltonian is strictly positive:*

$$\mathrm{Hess}\, H(\varphi) \ge \lambda \,\mathrm{Id}, \quad \forall \varphi \in \mathbb{R}^N. \tag{1.26}$$

Then (LSI) for ν holds with constant $\gamma \ge \lambda$.

For more general models, such as Gibbs measures of the type (1.1) in continuum scaling limit, or Hamiltonians without strict convexity, one may simplify by using decomposition of the entropy. Suppose ν is a probability distribution for spin configurations $(\varphi_1, \varphi_2) \in \mathbb{R}^{2N}$, then we have

$$\mathrm{Ent}_\nu[F(\varphi_1,\varphi_2)] = \mathbb{E}_\nu\big(\mathbb{E}_\nu(\Phi(F)|\varphi_2) - \Phi[\mathbb{E}_\nu(F|\varphi_2)]\big) + \mathbb{E}_\nu\big(\Phi[\mathbb{E}_\nu(F|\varphi_2)]\big) - \Phi[\mathbb{E}_\nu(F)] \tag{1.27}$$

where, in the above, $\mathbb{E}_\nu(\cdot|\varphi_2)$ refers to the conditional expectation with fixed $\varphi_2 \in \mathbb{R}^{\mathbb{N}}$. Suppose that we could estimate the first term (the relative entropy with respect to the conditional distribution) uniformly in φ_2, the

problem then reduces to the study of the 'renormalized entropy' on the distribution of spin configurations φ_2 after tracing out the fluctuations of φ_1. If $\nu = \nu_1 \otimes \nu_2$, the upper bound for the renormalized entropy is trivially given by the relative entropy for ν_2. Thus, if γ_1 and γ_2 denote the log-Sobolev constants for ν_1 and ν_2 respectively, the log-Sobolev inequality holds for ν with constant $\gamma = \min\{\gamma_1, \gamma_2\}$.

The second approach for the finite-volume LSI is the martingale method introduced by Lu and Yau [LY93] (see also [Led01-1, Section 5] for explanations), which provides a different way to tensorize the relative entropy. Suppose that we label the vertices in the finite subgraph Λ by $\{1, ..., N\}$, and let μ_k denote the conditional distribution on $\mathbb{R}^{N-k}$ given spin variables $\{\varphi_1, ..., \varphi_k\}$, $1 \le k \le N$. For any non-negative function $F : \mathbb{R}^N \to \mathbb{R}_+$, let $F_k = \mathbb{E}_{\mu_k}(F)$ and $F_0 = \mathbb{E}_\mu(F)$, the relative entropy (1.22) is decomposed as

$$\mathrm{Ent}_\mu(F) = \sum_{k=1}^{N} [\mathbb{E}_\mu(F_k \log F_k) - \mathbb{E}_\mu(F_{k-1} \log F_{k-1})] = \sum_{k=1}^{N} \mathbb{E}_\mu[\mathrm{Ent}_{\mu_{k-1}}(F_k)]. \tag{1.28}$$

If the log-Sobolev inequalities for all one-dimensional marginals are a priori assumed, then the right hand side of (1.28) is upper bounded by the sum of single-site Dirichlet forms of F_k, $k = 1, ..., N$. Thus, it suffices to bound them by the full Dirichlet forms of F. The difficulty of this step is to estimate the covariance between F and the partial derivatives of the Hamiltonian $\partial_k H$. The advantage of Lu-Yau's approach is that the proof relaxes the global convexity condition of the Hamiltonian to the convexity conditions only for one-dimensional marginal distributions. This will be particularly useful if the Gibbs measure is product on $\mathbb{R}^\Lambda$, see Section 2.1 for applications to the Ginzburg-Landau model.

As mentioned previously, this paper considers two types of stochastic dynamics in the thermodynamic limit $\Lambda \uparrow \mathbb{Z}^d$. However, for infinite-volume systems, our finite-volume argument fails due to the problem that the classical Sobolev inequalities are usually not satisfied in $\mathbb{R}^\infty$ (see [GZ03, Chapter 3]). To solve this issue, we may approximate $\mathbb{Z}^d$ by Λ, and study the asymptotic behavior of the finite-volume LSI constants. In some cases, this truncation gives no additional cost. For example, Aizenman and Holley [AH87] first proved that in spin systems with infinite-dimensional state space, the spectral gap for the Glauber generator is strictly positive if the Dobrushin-Shlosman uniqueness condition is satisfied. Later, Zegarlinski [Zeg92] and Strook-Zegarlinski [SZ92] proved the validity of log-Sobolev inequality, as well as Dobrushin-Shlosman mixing conditions for such models. As another example, the Gibbs measure for an ∞-temperature spin systems (i.e., when $\beta = 1/T = 0$) is a product measure, hence LSI holds uniformly in the size of Λ. Unfortunately, for many interesting statistical mechanics models with phase transitions, the correlation lengths are expected to blow up near criticality [Bax82, DMS97], and the tensorization property for the high-temperature measures does not hold. As a result, the finite-volume LSI constants are expected to diverge in the size of Λ and the uniform ergodicity of the infinite-volume dynamics fails.

To further explain the issue, consider the Glauber dynamics of Ising models on Λ_L. It is believed that in all dimensions $d \ge 2$, the inverse spectral gap of the generator (1.15) is of order-1 for $\beta < \beta_c$, polynomial in L for $\beta = \beta_c$, and exponential in L for $\beta > \beta_c$ (see [HH77, LF93, M99, WH97]). As a result, on the full integer lattice $\mathbb{Z}^d$, the exponential ergodicity observed in the high-temperature phase slows down to an algebraic rate of relaxation at the critical temperature. The situation is even worse for the Kawasaki dynamics of continuous spin fields on $\mathbb{Z}^d$, in which the conservation of particles leads to a negative correlation property. Namely, $\sum_{x\in\Lambda} \varphi_x = \text{const}$ implies $\mathrm{Cov}_\nu(\sum_{x\in\Lambda} \varphi_x, \sum_{x\in\Lambda} \varphi_x) = 0$, and henceforth $\mathrm{Cov}_\nu(\varphi_x, \varphi_y) \sim -L^{-d}$ for $x \ne y$. In high-temperature systems with strong spatial mixing properties, the relaxation of the Kawasaki dynamics is expected to be on the diffusive time scale $O(t^{-d/2})$ [CCR05, CM00, CMR02, LY93], which is much slower than the Glauber dynamics. Our main results in this article will focus on these types of polynomial rate of relaxation.

2 Finite-volume log-Sobolev inequalities

2.1 Martingale method and local central limit theorem

In this section, we derive quantitative bounds on the LSI constants for the Kawaskai dynamics (1.5) on cubes Γ_L. We will follow the strategy that relies on the aforementioned martingale method. However, the actual proof is more technical than the informal arguments discussed in Section 1.3, and the reason is twofold: (1) First, in the proof of Theorem 2.1.1, we need to consider the canonical measures $\nu_M^{\Gamma_L}$ supported on $\{\varphi \in \mathbb{R}^{\Gamma_L} | \sum_{x\in\Gamma_L} \varphi_x = M\}$. However, for the marginal entropy decomposition (1.28) of such measures, the extra constraints on the total number of particles complicate the calculation for single-site LSI constants; (2) Second, to bound the covariance between F and the derivatives of the global Hamiltonian $\sum_{x\in\Gamma_L} V'(\varphi_x)$, we need to further decouple $\nu_M^{\Gamma_L}$ into a family of multi-canonical measures with smaller supports: $\nu_M^{\Gamma_L} \mapsto \nu_{M_1}^{I_1} \otimes \cdots \nu_{M_n}^{I_n}$. Since this decoupling should be arbitrary, the bound on the local fluctuations should be uniform in the parameters $(M_1, ..., M_n)$. Fortunately, the ∞-temperature model (1.8) admits uniform (in λ) local central limit theorem (CLT) and large deviation estimates that solve the two problems.

Theorem 2.1.1 ([LPY02]). *Let $\gamma_{L,M}$ denote the LSI constant for the canonical measure $\nu_M^{\Gamma_L}$ (1.9), and $\gamma_L = \inf_{M\in\mathbb{R}} \gamma_{L,M}$. For any perturbative potential $V(x)$ with $\|\partial_x^2 U\|_\infty < \infty$, there exists a constant C depending only on U such that*

$$\frac{1}{\gamma_L} \le CL^2, \quad \forall L \ge 2. \tag{2.1}$$

Proof. Without loss of generality, we only prove the theorem for $d = 1$. According to [KL99, Section A.3.3], the result can be extended to higher dimensions. First, we focus on the log-Sobolev inequality for the single-site marginals of $\nu_M^{\Gamma_L}$, and inductively derive a recursive formula (relating γ_{L-1} and γ_L) using decomposition of entropy.

Let $L \ge 2$, $F : \mathbb{R}^{\Gamma_L} \mapsto \mathbb{R}$ be such that $\mathbb{E}_{\nu_M^{\Gamma_L}}(F^2) = 1$, and $\nu_{M,1}^{\Gamma_L}$ denote the marginal distribution of $\nu_M^{\Gamma_L}$ on the first site in Γ_L. Due to the fact that the total spins in Γ_L is fixed to be M, there exists a function $G(\varphi_1) = \sqrt{\mathbb{E}_{\nu_M^{\Gamma_L}}(F^2|\varphi_1)}$ such that $\mathbb{E}_{\nu_{M,1}^{\Gamma_L}}(G^2) = \mathbb{E}_{\nu_M^{\Gamma_L}}(F^2) = 1$. One can first prove that

$$\mathbb{E}_{\nu_{M,1}^{\Gamma_L}}[\Phi(G^2)] \lesssim_U \mathbb{E}_{\nu_{M,1}^{\Gamma_L}}[(\mathcal{L}_1 G)^2] \tag{2.2}$$

holds for any such normalized function G and $M \in \mathbb{R}$.

In fact, certain inequality for grand canonical measures, similar to (2.2), is trivial. This is because $\bar{\nu}_\lambda$ is a product measure and its single-site marginal has the probability density proportional to $\exp[-(x^2/2) - U(x)]$. Using integration by parts and elementary change of variables, one may upper bound the density by $100\exp(10\|U\|_\infty)$. The problem then reduces to proving LSI for the Lebesgue measure, which holds with an order-1 LSI constant. For canonical measures, we need to introduce several additional notations before estimating the density. Let $\{X_j^\lambda\}_{1\le j\le L}$ denote a family of i.i.d. random variables with common distribution: $g_\lambda(x)dx = Z(\lambda)^{-1}e^{\lambda x - V(x)}dx$. For this 1-dimensional measure, let $m_1(\lambda)$, $\sigma(\lambda)$, and $m_3(\lambda)$ denote the corresponding mean, variance, and centered third-order moment $\mathbb{E}(X_j^\lambda - m_1(\lambda))^3$, respectively. As a result, one has the local central limit theorem (CLT) for sufficiently large $L \gg L_0$:

$$\Big(f_{\lambda,L}(x) - (2\pi)^{-1/2}e^{-x^2/2}\big[1 - \frac{m_3(\lambda)x}{6\sigma(\lambda)^3\sqrt{L}}\big]\Big)^2 \lesssim L^{-2}, \quad \text{where } f_{\lambda,L}(x)dx \overset{d}{:=} \frac{\sum_{j=1}^L X_j^\lambda - m_1(\lambda)L}{\sigma(\lambda)\sqrt{L}}. \tag{2.3}$$

Moreover, the constant factor omitted in '$\lesssim$' is shown to be uniform in λ [LPY02, Theorem 5.2]. Thus, mimicking the grand canonical case, let $\lambda = \Upsilon[M/(L+1)]$ (recall Section 1.1 for definitions), the left-hand side of (2.2) is written as

$$\frac{\sqrt{L+1}}{\sqrt{L}} \int_{\mathbb{R}} G(x)^2 \log G(x)^2 g_\lambda(x) \frac{f_{\lambda,L}(\frac{\gamma_1(\lambda)-x}{\sigma\sqrt{L}})}{f_{\lambda,L+1}(0)} dx \tag{2.4}$$

Indeed, the denominator within the integral refers to the global constraint that $\sum_{1\le j\le L+1}\varphi_j = M$, and the numerator is the density for the fluctuations of $\{X_j^\lambda\}_{2\le j\le L+1}$ given the value $\varphi_1 = x$ of the first spin. When L is smaller than the threshold for local CLT, elementary calculations ([LPY02, Lemma 5.6]) yield

$$\frac{\sqrt{L+1}}{\sqrt{L}} g_\lambda(x) \frac{f_{\lambda,L}(\frac{\lambda-x}{\sigma\sqrt{L}})}{f_{\lambda,L+1}(0)} \asymp_{L_0} e^{-\frac{1}{2}(1+L^{-1})(x-\lambda)^2}, \tag{2.5}$$

where the equivalence '$\asymp_{L_0}$' is uniform in λ but dependent on L_0. When $L \gg L_0$, we apply (2.3) to show that $f_{\lambda,L}(0)$ is bounded above and below by a uniform constant. To be more specific, setting $\theta = \Upsilon[(1+L^{-1})m_1(\lambda) - xL^{-1}]$ gives the following

$$\frac{\sqrt{L+1}}{\sqrt{L}} g_\lambda(x) \frac{f_{\lambda,L}(\frac{\lambda-x}{\sigma\sqrt{L}})}{f_{\lambda,L+1}(0)} \asymp e^{-\frac{1}{2}(x-\lambda^2)+L[\log Z(\lambda)-\log Z(\theta)-(\lambda-\theta)m_1(\theta)]}. \tag{2.6}$$

In conclusion, (2.5) and (2.6) together implies the claim (2.2), since the 1-dimensional measure is shown to be equivalent to a small perturbation of the Gaussian distribution. The strict convexity of the quadratic Gaussian potential allows the use of the Bakry-Emery criterion.

By (1.27), we decompose the relative entropy as

$$\mathrm{Ent}_{\nu_M^{\Gamma_L}}(F^2) = \mathbb{E}_{\nu_M^{\Gamma_L}}[\Phi(F^2)] = \mathbb{E}_{\nu_M^{\Gamma_L}}\Big[F^2 \log \frac{F^2}{G^2}\Big] + \mathbb{E}_{\nu_{M,1}^{\Gamma_L}}[\Phi(G^2)] \tag{2.7}$$

To bound the first term on the right-hand side of (2.7), we argue by induction and assume that (2.1) holds for Γ_{L-1}, then

$$\begin{aligned}\mathbb{E}_{\nu_M^{\Gamma_L}}[F^2\log(F^2/G^2)] &= \mathbb{E}_{\nu_{M,1}^{\Gamma_L}}\big([\mathbb{E}_{\nu_{M-\varphi_1}^{\Gamma_{L-1}}}\big(\Phi(F^2/G^2)\big]\cdot G^2\big) \\ &\le \frac{1}{\gamma_{L-1}}\mathbb{E}_{\nu_{M,1}^{\Gamma_L}}\big(\mathcal{D}_{\nu_{M-\varphi_1}^{\Gamma_{L-1}}}(F/G)\cdot G^2\big) \le \frac{1}{\gamma_{L-1}}\mathcal{D}_{\nu_M^{\Gamma_L}}(F).\end{aligned} \tag{2.8}$$

For the second term, we apply (2.2) and calculate

$$\mathbb{E}_{\nu_{M,1}^{\Gamma_L}}[\Phi(G^2)] \lesssim_U \mathbb{E}_{\nu_{M,1}^{\Gamma_L}}\Big(L\sum_{j=2}^{L}\mathbb{E}_{\nu_M^{\Gamma_L}}[(T^{j,j+1}F)^2|\varphi_1] - \mathrm{Cov}_{\nu_M^{\Gamma_L}}\Big[F^2, \frac{1}{L-1}\sum_{j=2}^{L}V'(\varphi_j)\Big|\varphi_1\Big]\Big)^2. \tag{2.9}$$

In particular, the first part in the above resembles a Kawasaki Dirichlet form. The second part is the conditional covariance between F^2 and the average of the gradient ∇V on Γ_{L-1}, given φ_1 at the first vertex of Γ_L. Suppose that one could upper bound the conditional covariance by the product of the desired Dirichlet form and a coefficient involving γ_{L-1}^{-1}. Then, combining (2.8) and (2.9) yields a recursive formula for γ_L and γ_{L-1}. This is sufficient to imply that $\gamma_L \lesssim_U L^2$, which concludes the proof of the theorem.

To maintain consistency throughout the article, we provide only a brief comment on the proof of the covariance bounds.

Proposition 2.1.2. *For any $L \ge 2$, $M \in \mathbb{R}$, and $F \in L^2(\nu_M^{\Gamma_L})$ such that $\mathbb{E}_{\nu_M^{\Gamma_L}}(F^2) = 1$, we have*

$$\mathrm{Cov}_{\nu_M^{\Gamma_L}}\Big[F^2, \frac{1}{L}\sum_{j=1}^{L}V'(\varphi_j)\Big]^2 \lesssim_U \frac{1}{L\gamma_L}\mathcal{D}_{\nu_M^{\Gamma_L}}(F). \tag{2.10}$$

Moreover, for a fixed $\epsilon \in (0,1)$, there exists $L_0 \gg 2$ (depending only on ϵ and U) such that

$$\mathrm{Cov}_{\nu_M^{\Gamma_L}}\Big[F^2, \frac{1}{L}\sum_{j=1}^{L}V'(\varphi_j)\Big]^2 \lesssim_U \Big(L + \frac{\epsilon}{L\gamma_L}\Big)\mathcal{D}_{\nu_M^{\Gamma_L}}(F) \tag{2.11}$$

holds for any $L \ge L_0$, $M \in \mathbb{R}$, and normalized L^2 function F.

The first covariance bound (2.10) only proves our theorem for small L. In fact, (2.10) and the entropy decomposition (2.7) deduce the *1st recursive formula*:

$$\frac{1}{\gamma_L} \leq C_0 L + \frac{(1 + C_0 L^{-1})}{\gamma_{L-1}}. \tag{2.12}$$

Since $\gamma_2^{-1} \leq C_0$ (recall the single-site case (2.2)), we have $\gamma_L^{-1} \leq C_0^L + C_0 L^2$. The leading order is exponentially large, which is not sufficient for the full regime: $L \geq 2$. The second covariance bound (2.11) leads to the *2nd recursive formula*:

$$\frac{1}{\gamma_L} \leq C_0 L + \frac{(1 + \epsilon L^{-1})}{\gamma_{L-1}}, \quad L \geq L_0. \tag{2.13}$$

Due to the improvement of the parameter (C_0 replaced by ϵ), we can neglect the exponential error ϵ^L and conclude that $\gamma_L^{-1} \lesssim_U L^2$.

In order to prove Proposition 2.1.2 in the small-L case, we consider the entropy inequality (see, e.g., [Var16, Lemma 4.9]):

$$\mathbb{E}_{\nu_M^{\Gamma_L}}(F^2 H) \leq \log \mathbb{E}_{\nu_M^{\Gamma_L}}(e^H) + \mathrm{Ent}_{\nu_M^{\Gamma_L}}(F). \tag{2.14}$$

In our context, the function H refers to the averaging of the gradient ∇V. Using Varadhan's lemma (see [Var16, Theorem 2.6], or [LPY02, Lemma 6.1]), the term $\log \mathbb{E}_{\nu_M^{\Gamma_L}}(e^H)$ is upper bounded by a constant factor involving only V. For the large-L case, we define spatial decompositions $\Gamma_L = \cup_{i=1}^n I_i$ and multiscale decomposition $M = \sum_{i=1}^n M_i$. The fluctuations of the spins in the large cube Γ_L is then factorized into a multi-canonical measure $\nu_{M_1}^{I_1} \otimes \cdots \nu_{M_n}^{I_n}$. By taking entropy inequalities (2.14) for all cubes $\{I_i, 1 \leq i \leq n\}$ and averaging over all possible sums $M = \sum_{i=1}^n M_i$, the desired sharper bound (2.11) follows. □

2.2 Renormalization group approach and the Polchinski equation

Recall Section 1.2, phase transition phenomena of the Ising model is also characterized by the critical behavior of physical observables such as the spontaneous magnetization m_β, spin-spin correlation function $\langle \sigma_0 \sigma_x \rangle_\beta$, susceptibility χ_β, etc. Moreover, there exists quantitative descriptions of such critical phenomena using several critical exponents, and the scaling hypothesis imposes connections between these exponents, see [DMS97] for related discussions. An informal way to justify the scaling hypothesis for the critical Ising model is Kadanoff's block-spin renormalization [Kad66]. Since the high-temperature phase is disordered, spins σ_x and σ_y with distance $|x-y|$ much larger than the correlation length ξ_β tend to align independently. However, at critical temperature, spins fluctuate over any scale, and a finite correlation length no longer exists. According to Kadanoff's idea, if one rescales the critical Ising model by replacing a block of neighboring sites by one site exhibiting the dominant spin, the statistics of the model are left invariant under a companion change of the coupling parameters. Repeating this argument leads to a flow map on the space of all coupling parameters, and hence the critical Ising model is interpreted as a renormalization group (RG) fixed point.

In certain critical lattice models, the long-range correlation property is expected to be described by the Gaussian free field [WK74]. Motivated by this idea, Bauerschmidt, Brydges, and Slade [BBS19] developed another approach to continuously renormalize lattice φ^4 and Ising models. Instead of considering the Lebesgue measure $d\varphi$ on $\mathbb{R}^\Lambda$ as reference, one regards spin field distribution as fluctuations over a given DGFF. By continuously decomposing the covariance of the coupled DGFF, the parameters of the system are required to evolve continuously, and this can be thought of as a RG flow. In a recent expository paper, Bauerschmidt, Bodineau, and Dagallier [BBD24] presented an overview of this approach to the study of various stochastic dynamics. In particular, they used the decomposition of covariance to obtain a continuous family of entropy decompositions (recall (1.27)). The invariance under this RG flow map is described by the Polchinski equation, and this leads to the multiscale Bakry-Emery criterion for the log-Sobolev inequality of the Gibbs measure. The following contexts in this section comprise the overview of [BD24-1] for the (near) critical Ising model. Using similar theoretical frameworks, LSIs for the continuum sine-Gordon model and the lattice φ_2^4 and φ_3^4 models were derived in [BB21, BD24-2].

In Bauerschmidt-Dagallier [BD24-1], they considered a more general ferromagnetic critical Ising model on Λ, with arbitrary external field h and free boundary condition:

$$\mu^{\Lambda}_{\beta_c,h}(\sigma) \propto e^{-\frac{\beta}{2}(\sigma,A\sigma)+(h,\sigma)} \tag{2.15}$$

where $(\cdot,\cdot)$ denotes the standard inner product on $\mathbb{R}^{\Lambda}$, and A is any positive definite matrix with spectral radius $\|A\| \leq 1$ (if A is the adjacency matrix associated to the graph Λ, then we recover the nearest-neighbor interaction (1.13)). For the log-Sobolev inequality associated to the measure (2.15), we consider the standard Glauber-Dirichlet form:

$$\mathrm{D}_{\mu^{\Lambda}_{\beta_c,h}}(F) = \frac{1}{2}\sum_{x\in\Lambda}\mathbb{E}_{\mu^{\Lambda}_{\beta_c,h}}\big[(\nabla_x F)^2\big] = \frac{1}{2}\sum_{x\in\Lambda}\mathbb{E}_{\mu^{\Lambda}_{\beta_c,h}}\big[\big(F(\sigma)-F(\sigma^x)\big)^2\big]. \tag{2.16}$$

Recall that we have denoted σ^x as the spin configuration obtained by flipping σ at $x \in \Lambda$. According to the definition of Glauber dynamics (see section 1.2), the uniform boundedness of the jump rates $c(x,\sigma)$ implies that the two Dirichlet forms (1.20) and (2.16) are essentially equivalent. The main result for the LSI constant contains the 0-field magnetic susceptibility χ_β (at any temperature $\beta \in [0,\beta_c]$) of the Ising model, which is defined by

$$\chi_\beta = \sup_{x\in\Lambda}\sum_{y\in\Lambda}\mathbb{E}_{\mu^{\Lambda}_{\beta}}(\sigma_0\sigma_x) \tag{2.17}$$

Since the graph is always fixed, we omit Λ in the following for brevity of notation.

Theorem 2.2.1 ([BD24-1]). *The log-Sobolev constant $\gamma_{\beta_c,h}$ for the measure (2.15) and Dirichlet form (2.16) satisfies the bound*

$$\frac{1}{\gamma_{\beta_c,h}} \leq \frac{1}{2} + \int_0^{\beta_c} e^{2\int_0^t \chi_s ds} dt. \tag{2.18}$$

Proof. Since Ising spins are bounded, the measure (2.15) is left invariant if we replace A by any matrix $A+\alpha$, $\alpha \in \mathbb{R}$. Thus, fix any $\alpha > \beta_c$, we consider the interpolation of coupling matrices between $\alpha\,\mathrm{Id}$ and $\beta_c A + (\alpha-\beta_c)$:

$$A_t = tA + (\alpha - t), \quad C_t = A_t^{-1}, \quad t \in [0,\beta_c]. \tag{2.19}$$

In particular, at $t=\beta_c$, A_{β_c} is the coupling for our Ising model. The main idea of the renormalization is that at any $t \in [0,\beta_c)$, we extract a DGFF distribution from $\mu_{\beta_c,h}$ with the covariance matrix given by $C_{\beta_c} - C_t$. This is slightly different from the covariance decomposition considered in [BB21], where C_t is parametrized by $t \in [0,\infty]$ and $C_0 = 0$. In fact, they are equivalent since (2.19) is implied by a change of variable $t \mapsto \beta_c(1-e^{-t})$, and the replacement $C_t \mapsto C_t - C_0$.

For any $t \in [0,\beta_c)$, we define the renormalized potential for continuous Gaussian spin fields φ by

$$V_t(\varphi) = -\log\sum_{\sigma\in\{\pm1\}} e^{-\frac{1}{2}(\sigma-\varphi,C_t^{-1}(\sigma-\varphi))+(h,\sigma)}. \tag{2.20}$$

As a result, we obtain the renormailzed measure (i.e., the DGFF) on $\mathbb{R}^{\Lambda}$ given by

$$\nu_{t,\beta_c}(d\varphi) \propto e^{-\frac{1}{2}(\varphi,(C_{\beta_c}-C_t)^{-1}\varphi)-V_t(\varphi)}d\varphi, \tag{2.21}$$

and the fluctuation measure (conditional on a given field φ) for the Ising spins is expressed by

$$\mu_{t,h+C_t^{-1}\varphi}(\sigma|\varphi) \propto e^{-\frac{1}{2}(\sigma,C_t^{-1}\sigma)+(h+C_t^{-1}\varphi,\sigma)}. \tag{2.22}$$

The full critical Ising expectation (2.15) is now decomposed into

$$\mathbb{E}_{\mu_{\beta_c,h}}[F] = \mathbb{E}_{\nu_{t,\beta_c}}[\mathbb{E}_{\mu_{t,h+C_t^{-1}\varphi}}(F)] \tag{2.23}$$

The factorization identity (2.23) is fairly easy to check. Indeed, we apply convolution properties of the Gaussian integrals and compute

$$\begin{aligned}\mathbb{E}_{\mu_{\beta_c,h}}[F] &\propto \sum_{\sigma\in\{\pm1\}^\Lambda} e^{-\frac{1}{2}(\sigma,C_{\beta_c}^{-1}\sigma)+(h,\sigma)}F(\sigma)\\ &\propto \sum_{\sigma\in\{\pm1\}^\Lambda}\int_{\mathbb{R}^\Lambda} d\varphi e^{-\frac{1}{2}(\varphi,C_t^{-1}\varphi)}e^{-\frac{1}{2}(\sigma-\varphi,C_t^{-1}(\sigma-\varphi))}e^{(h,\sigma)}F(\sigma)\\ &= \int_{\mathbb{R}^\Lambda} d\varphi e^{-\frac{1}{2}(\varphi,C_t^{-1}\varphi)-V_t(\varphi)}\Big(\frac{1}{e^{-V_t(\varphi)}}\sum_{\sigma\in\{\pm1\}^\Lambda} e^{-\frac{1}{2}(\sigma-\varphi,C_t^{-1}(\sigma-\varphi))}e^{(h,\sigma)}F(\sigma)\Big)\end{aligned} \tag{2.24}$$

The bracket term in (2.24) corresponds to the Polchinski flow map $\mathbf{P}_{0,t}$ on the space of non-negative functions, see [BBD24] for definitions and the derivation of the Polchinski equation for the renormalized potential.

Let $F:\{\pm1\}^\Lambda\to\mathbb{R}$ be a fixed observable. According to (1.27), we apply the RG flow map at $t=0$ and decompose the relative entropy by:

$$\mathrm{Ent}_{\mu_{\beta_c,h}}(F)=\mathbb{E}_{\nu_{0,\beta_c}}(\mathrm{Ent}_{\mu_{0,h+C_0^{-1}\varphi}}(F))+\mathrm{Ent}_{\nu_{0,\beta_c}}(\mathbb{E}_{\mu_{0,h+C_0^{-1}\varphi}}(F)). \tag{2.25}$$

Since $C_0^{-1}=\alpha$, the measure $\mu_{0,h+C_0^{-1}\varphi}$ is the ∞-temperature Ising measure with external field $h+\alpha\varphi$. Similar to the Kawasaki dynamics of the (product) Ginzburg-Landau model, the log-Sobolev inequality for this part is uniform in Λ, h, and φ with LSI constant $\gamma=2$, see [Led01-2, LSC97]. Thus, the first term in (2.25) is upper bounded by

$$\mathbb{E}_{\nu_{0,\beta_c}}(\mathrm{Ent}_{\mu_{0,h+C_0^{-1}\varphi}}(F))\le\mathbb{E}_{\nu_{0,\beta_c}}\mathrm{D}_{\mu_{0,h+C_0^{-1}\varphi}}(\sqrt{F})=\mathrm{D}_{\mu_{\beta_c,h}}(\sqrt{F}). \tag{2.26}$$

For the second term in (2.25), we revisit the multiscale Bakry-Emery criterion developed in the seminal work [BB21]. The proof is similar to the one for Bakry-Emery criterion, except that the Langevin dynamics is replaced by the Polchinski equation, see also [BBD24, Section 3] for details. We remark that if the original probability measure is strictly log-concave, then the Polchinski flow preserves the convexity of the Hamiltonian, and the classical Bakry-Emery criterion is recovered.

Theorem 2.2.2 (multiscale Bakry-Emery criterion)**.** *Suppose $(\dot\lambda_t)_{t\in[0,\beta_c]}$ is a family of real numbers, and $(C_t)_{t\in[0,\beta_c]}$ is the continuous covariance decomposition such that*

$$C_{\beta_c}-C_0=\int_0^{\beta_c}\dot C_t dt,\quad \dot C_t\,\mathrm{Hess}\,V_t(\varphi)\dot C_t-\frac{1}{2}\ddot C_t\ge\dot\lambda_t\dot C_t,\quad\forall t\in[0,\beta_c]. \tag{2.27}$$

Then one obtains LSI with constant γ satisfying

$$\frac{1}{\gamma}\le\|\dot C_0\|\int_0^{\beta_c}e^{-2\int_0^t\dot\lambda_s ds}dt \tag{2.28}$$

In the statement of the above theorem, we have parametrized the covariance decomposition up to the critical temperature to accommodate the Ising model. To find a suitable family $(\dot\lambda_t)_{t\in[0,\beta_c]}$, we compute

$$\mathrm{Hess}\,V_t(\varphi)=C_t^{-1}-C_t^{-1}\Sigma_t(h+C_t^{-1}\varphi)C_t^{-1} \tag{2.29}$$

where $\Sigma_t(*)$ refers to the covariance matrix for the fluctuation measure (2.22) with any external field $h=*$:

$$\Sigma_t(*)=\Big(\mathbb{E}_{\mu_{t,*}}(\sigma_x\sigma_y)-\mathbb{E}_{\mu_{t,*}}(\sigma_x)\mathbb{E}_{\mu_{t,*}}(\sigma_y)\Big)_{x,y\in\Lambda}. \tag{2.30}$$

By FKG inequality, the entries of $\Sigma_t(*)$ are non-negative. In addition, due to the work of Ding, Song, and Sun [DSS22], we know that spin-spin correlation functions of the ferromagnetic Ising models are maximized when the external field is zero. Using Perron-Frobenius theorem, there exists $Y_0 \in \mathbb{R}^\Lambda$ such that

$$\|\Sigma_t(*)\| = Y_0^T \Sigma_t(*) Y_0 \le Y_0^T \Sigma_t(0) Y_0 \le \|\Sigma_t(0)\| \le \chi_t. \tag{2.31}$$

In particular, the first inequality is a consequence of Ding-Song-Sun's inequality. Combining (2.29) and (2.31), one can calculate and show that

$$\dot{C}_t \operatorname{Hess} V_t(\varphi) \dot{C}_t - \frac{1}{2}\ddot{C}_t \ge -\chi_t \dot{C}_t. \tag{2.32}$$

Theorem 2.2.2 then shows that the second term in (2.25) is upper bounded by

$$\operatorname{Ent}_{\nu_{0,\beta_c}}(\mathbb{E}_{\mu_{0,h+C_0^{-1}\varphi}}(F)) \le \frac{2}{\alpha^2}\Big(\int_0^{\beta_c} e^{2\int_0^t \chi_s ds} dt\Big) \mathrm{D}_{\nu_{0,\beta_c}}\big(\sqrt{\mathbb{E}_{\mu_{0,h+C_0^{-1}\varphi}}(F)}\big). \tag{2.33}$$

The remaining steps of the proof are all about replacing the Dirichlet form in the right-hand side by the standard one for F (2.16). This is solved by calculating the covariance between F and single spins σ_x under the measure $\mu_{0,h+\alpha\varphi}$. The resulting upper bound generates an extra α^2 parameter, which concludes the proof of (2.18), see [BD24-1] for details. □

Remark 2.2.3. Due to the second Griffiths inequality (see [FV18, Exercise 3.9]), the susceptibility χ_s is an increasing function in $s \in [0,\infty)$. Bauerschmidt and Dagallier's result (replace β_c by any β in the proof of Theorem 2.2.1) then implies the trivial bound

$$\frac{1}{\gamma_{\beta,h}} \le \frac{1}{2} + \beta e^{2\beta\chi_\beta}. \tag{2.34}$$

This shows that, in the entire high-temperature regime $\beta \in [0,\beta_c)$, the LSI constant is uniform in the size of the graph Λ, and hence the thermodynamic limit of the Glauber dynamics should exhibit an exponential rate of relaxation.

Near criticality, χ_β diverges algebraically fast in $(\beta_c - \beta)^{-1}$, as characterized by the critical exponents of the Ising model. (2.18) shows that if the susceptibility satisfies an additional mean-field bound, the critical log-Sobolev constant $\gamma_{\beta_c,h}$ should be polynomial. In fact, for the Curie-Weiss model on any graph, we have

$$\chi_\beta \le \frac{1}{\beta_c - \beta} \tag{2.35}$$

for any $\beta < \beta_c$, see [BBS19, Chapter 1]. If we consider Ising models (1.13) on $\Lambda_L \subset \mathbb{Z}^d$ with nearest-neighbor interactions, the mean-field behavior is expected to appear in $d \ge 5$ [Aiz82, AG83, CJN21]. One can intuitively imagine that the structure of the high-dimensional integer lattice should be similar to a complete graph. As a consequence, we have the following polynomial bound for the critical LSI constant in Λ_L, which is crucial in the proof of our main result (0.2).

Corollary 2.2.4. *Consider the nearest-neighbor Ising model (1.13) at $d \ge 5$. For $D > 1/2$, the high-temperature susceptibility and the critical log-Sobolev constant in Λ_L satisfy the bounds:*

$$\chi_\beta \le \frac{D}{\beta_c - \beta + L^{-2}}, \quad \beta < \beta_c; \quad \frac{1}{\gamma^\tau_{\beta_c}} \le \frac{\beta_c}{2D-1}(L^2\beta_c + 1)^{2D-1}. \tag{2.36}$$

Theorem 2.18 is slightly stronger than the above finite-volume bounds with an arbitrary boundary condition since it is valid under any external fields. In the statement of Theorem 0.0.2, the constant η should be understood as $\eta = 2(2D-1)$.

In lower dimensions $d = 2, 3, 4$, the polynomial estimate for the finite-volume critical LSI constant is still unknown. For the inverse spectral gap at critical temperature, Lubetzky and Sly [LS12] established a polynomial estimate on Λ_L at $d = 2$. Using conformal symmetry (RSW theory) and the method of block spin dynamics, Lubetzky and Sly proved their result by establishing a spatial mixing property of an entire face of the boundary. This compensates for the loss of the strong mixing properties of single spins at critical temperature. For dimensions $d = 3, 4$, no rigorous result is known at this point.

3 Polynomial rate of relaxations

3.1 Finite-volume approximation

We now return to the proof of Theorem 0.0.1. Heuristically speaking, the connections between the dynamics and the heat equations in $\mathbb{R}^d$ should lead to a $O(t^{-d/2})$ rate of convergence for the diffusion process (1.3). The logarithmic correction term $\log(2+t)$ suggests that (0.1) is not optimal and could be improved by using different methods. Since the LSI constant is known to decay polynomially fast to zero (2.1), quantitative estimates only work in finite volume and we have to approximate the dynamics on $\mathbb{Z}^d$ by those in large cubes $\Gamma_L \subset \mathbb{Z}^d$, $L \gg 2$. A natural strategy is to truncate $\mathbb{Z}^d$ by cubes with side-length $L = L(t)$ growing at a certain algberaic rate. This is considered in [JLQY99] for the zero-range processes. For continuous spin fields, multi-canonical measures are supported on hypersurfaces in $\mathbb{R}^{\Gamma_{L(t)}}$. As a result, to further decouple the dynamics on $\Gamma_{L(t)}$, we have to deal with a family of 'singular' Radon-Nikodym derivatives that are also growing in time. To reduce the complexity of computations, we consider truncations with L to be arbitrarily large (i.e., independent of t) but fixed. Assume that the convergence rate $O([\log(2+t)]t^{-d/2})$ holds for any L, the result will then follow from taking the thermodynamic limit $L \to \infty$.

Proposition 3.1.1. *Fix $d \geq 1$, $\omega \in \mathbb{R}$, and a local function $F \in L^2(\nu_\omega^{\Gamma_L})$ with finite triple norm. Denote by $\mathcal{P}_t^L$ the Markov semigroup associated to $\mathcal{L}_L$. There exists a constant $C_0 = C_0(d, \omega, V)$, independent of L, such that*

$$\mathrm{Var}_{\nu_\omega^L}(\mathcal{P}_t^L F) \leq \frac{C \,|||\, F \,|||^2 \log(2+t)}{(1+t)^{d/2}}. \tag{3.1}$$

Proof of Theorem 0.0.1. Without loss of generality, we may assume $F \in L^2(\nu_\omega)$ is means-zero and henceforth $\mathrm{Var}_{\nu_\omega^{\Gamma_L}}(\mathcal{P}_t F) = \mathbb{E}_{\nu_\omega}[(\mathcal{P}_t F)^2]$. For any $L \geq 1$, let $\mathfrak{F}_L$ denote the σ-algebra generated by $\{\varphi_x, x \in \Gamma_L\}$. By Cauchy-Schwarz inequality,

$$\mathbb{E}_{\nu_\omega}[(\mathcal{P}_t F)^2] \leq 2\mathbb{E}_{\nu_\omega}\{[\mathcal{P}_t \mathbb{E}_{\nu_\omega}(F|\mathfrak{F}_L)]^2\} + 2\mathbb{E}_{\nu_\omega}\{[\mathcal{P}_t F - \mathcal{P}_t \mathbb{E}_{\nu_\omega}(F|\mathfrak{F}_L)]^2\}. \tag{3.2}$$

Since the infinite-volume invariant measure ν_ω and semigroup $\mathcal{P}_t$ are approximated by $\nu_\omega^{\Gamma_L}$ and $\mathcal{P}_t^L$ respectively, we have

$$\begin{aligned}\mathbb{E}_{\nu_\omega}\{[\mathcal{P}_t \mathbb{E}_{\nu_\omega}(F|\mathfrak{F}_L)]^2\} &= \lim_{L\to\infty} \mathbb{E}_{\nu_\omega^{\Gamma_L}}\{[\mathcal{P}_t^L \mathbb{E}_{\nu_\omega}(F|\mathfrak{F}_L)]^2\} \\ &\leq \frac{C \,|||\, \mathbb{E}_{\nu_\omega}(F|\mathfrak{F}_L) \,|||^2 \log(2+t)}{(1+t)^{d/2}} \leq \frac{C \,|||\, F \,|||^2 \log(2+t)}{(1+t)^{d/2}},\end{aligned} \tag{3.3}$$

where, in the second line, we applied Proposition 3.1.1. Since our calculation holds with any $L \geq 1$, the proof is therefore concluded by inserting (3.3) back into (3.2) and taking the thermodynamic limit $L \to \infty$. □

Before we begin with the proof of the convergence in Γ_L, we recall the covariance bound for the proof of the log-Sobolev inequality (see Proposition 2.1.2). For large $L \gg 1$, we need to decompose the canonical measure by $\nu_M^{\Gamma_L} \mapsto \nu_{M_1}^{I_1} \otimes \cdots \otimes \nu_{M_n}^{I_n}$, followed by a suitable average over all possible vectors $\mathbf{M} = (M_1, ..., M_n)$ such that $M = \sum_{i=1}^n M_i$. However, in each smaller part I_i, the non-negative density $F^2 = d\nu_{M_i}^{I_i} \mathcal{P}_t^{I_i} / d\nu_{L,M}$ considered in the covariance (2.11) is, in fact, infinite. This is because the Kawasaki dynamics is conservative

and preserves the total spin vector $\mathbf{M}$. Due to the fact that $\{(\varphi_x)_{x\in\Gamma_L}|\sum_{x\in I_i}\varphi_x = M_i, 1\le i\le n\}$ has Lebesgue measure zero in subspace $\{(\varphi_x)_{x\in\Gamma_L}|\sum_{x\in\Gamma_L}\varphi_x = M\}$, the time-$t$ distribution $\nu_{M_i}^{I_i}\mathcal{P}_t^{I_i}$ is singular with respect to $\nu_M^{\Gamma_L}$.

To overcome this difficulty in the forthcoming covariance bound, we need to introduce a non-conservative Glauber perturbation to the Kawasaki semigroup. For any $\varepsilon>0$, denote by $\mathcal{L}_{\Gamma_L}^{G,\varepsilon}$ the generator defined by

$$\mathcal{L}_{\Gamma_L}^{G,\varepsilon} = \frac{\varepsilon}{2}\sum_{x\in\Gamma_L}\partial_x^2 - \frac{\varepsilon}{2}\sum_{x\in\Gamma_L}[\Upsilon(\omega) - V'(\varphi_x)]\partial_x. \tag{3.4}$$

From the choice of the drift term, we see that $\nu_\omega^{\Gamma_L}$ is invariant for the diffusion process with perturbative generator $\mathcal{L}_{\Gamma_L}^{\varepsilon} = \mathcal{L}_{\Gamma_L} + \mathcal{L}_{\Gamma_L}^{G,\varepsilon}$. Let $\mathcal{P}_t^{\varepsilon,L}$ denote the semigroup associated with $\mathcal{L}_{\Gamma_L}^{\varepsilon}$. We now show that for a fixed time $t\ge 0$, the Glauber perturbation leads to an L^2-error only up to order-ε.

Lemma 3.1.2. *Fix $\varepsilon>0$, $t\ge 0$, and a local function $F\in L^2(\nu_\omega^{\Gamma_L})$ with finite triple norm. There exists a constant $C'=C'(d,\omega,V)$ such that*

$$\mathbb{E}_{\nu_\omega}[(\mathcal{P}_t^{\varepsilon,L}F - \mathcal{P}_t^L F)^2] \le C'\,|||\,F\,|||^2\,\varepsilon t(1+t). \tag{3.5}$$

Proof. According to backward Kolmogorov equation, and the fact that $\mathcal{P}_0^{\varepsilon,L} = \mathcal{P}_0^L = \mathrm{Id}$, we have

$$\begin{aligned}
&\mathbb{E}_{\nu_\omega}[(\mathcal{P}_t^{\varepsilon,L}F - \mathcal{P}_t^L F)^2]\\
&= \int_0^t \Big(2\mathbb{E}_{\nu_\omega}[(\mathcal{P}_s^{\varepsilon,L}F - \mathcal{P}_s^L F)(\mathcal{L}_{\Gamma_L}^{G,\varepsilon}\mathcal{P}_s^{\varepsilon,L}F)] + 2\mathbb{E}_{\nu_\omega}[(\mathcal{P}_s^{\varepsilon,L}F - \mathcal{P}_s^L F)\mathcal{L}_{\Gamma_L}(\mathcal{P}_s^{\varepsilon,L}F - \mathcal{P}_s^L F)]\Big)ds\\
&\le -\int_0^t 2\mathbb{E}_{\nu_\omega}[(\mathcal{P}_s^L F)(\mathcal{L}_{\Gamma_L}^{G,\varepsilon}\mathcal{P}_s^{\varepsilon,L}F)]ds.
\end{aligned} \tag{3.6}$$

Recalling the definition of the Dirichlet form (1.11), we have removed the negative terms inside the integral. Using the results $\|\mathcal{L}_{\Gamma_L}^{G,\varepsilon}\| = O(\varepsilon)$ and $\|\mathcal{P}_s^{\varepsilon,L} - \mathcal{P}_s^L\| = O(s\varepsilon e^{s\varepsilon})$ from semigroup theory (see, for instance, [Yos80]), the order of the right-hand side of (3.6) is upper bounded by

$$-\int_0^t 2\mathbb{E}_{\nu_\omega}[(\mathcal{P}_s^L F)(\mathcal{L}_{\Gamma_L}^{G,\varepsilon}\mathcal{P}_s^L F)]ds \le \frac{\varepsilon}{2}\int_0^t\sum_{x\in\Gamma_L}\mathbb{E}_{\nu_\omega}[(\partial_x\mathcal{P}_s^L F)^2]ds \le C'\,|||\,F\,|||^2\,\varepsilon t(1+t). \tag{3.7}$$

The details are left as an exercise (or see [LY03]), which follows from applying the Cauchy inequality to the integrand. □

For any fixed but large time $T\gg 1$, take

$$\varepsilon = \frac{1}{(1+T)^{2+\frac{d}{2}}}. \tag{3.8}$$

Within this definition, the preceding lemma implies that Proposition 3.1.1 is equivalent to the following bound for the perturbative dynamics:

$$\mathrm{Var}_{\nu_\omega^{\Gamma_L}}(\mathcal{P}_T^{\varepsilon,L}F) \le \frac{C\,|||\,F\,|||^2\log(2+T)}{(1+T)^{d/2}}. \tag{3.9}$$

Due to complexity, we present the proof of (3.9) in the next subsection. The idea also applies to the Ising case, except for the local function chosen to be the single spin variable $F(\sigma)=\sigma_0$, which makes the calculations easier, see Section 3.4.

3.2 Proof of Claim 3.9

Step 1. Dynamical decoupling.

Without creating extra confusion, we sometimes write $\mathcal{P}_t$ for $\mathcal{P}_t^{\varepsilon,L}$ in the upcoming calculations. Given the infinite temporal regime $[1,\infty)$, we construct a partition by inductively taking $t_0=1$ and $t_n>2t_{n-1}$, $n\geq 1$. Let $T\gg 1$ be arbitrarily large but fixed, denote by $n(T)$ the largest integer such that $t_{n(T)-1}<T$, and redefine $t_{n(T)}:=T$. By backward Kolmogorov's equation, we may compute

$$\begin{aligned}
&(1+T)^{\frac{d+2}{2}}\mathbb{E}_{\nu_\omega}[(\mathcal{P}_T F)^2]-2^{\frac{d+2}{2}}\mathbb{E}_{\nu_\omega}[(\mathcal{P}_1 F)^2]\\
&\quad=\sum_{j=0}^{n(T)}\Big\{(1+t_{j+1})^{\frac{d+2}{2}}\mathbb{E}_{\nu_\omega}[(\mathcal{P}_{t_{j+1}}F)^2]-(1+t_j)^{\frac{d+2}{2}}\mathbb{E}_{\nu_\omega}[(\mathcal{P}_{t_j}F)^2]\Big\}\\
&\quad=\sum_{j=0}^{n(T)}\int_{t_j}^{t_{j+1}}\partial_t\Big\{(1+t)^{\frac{d+2}{2}}\mathbb{E}_{\nu_\omega}[(\mathcal{P}_t F)^2]\Big\}dt\\
&\quad=\sum_{j=0}^{n(T)}\Big\{\frac{d+2}{2}\int_{t_j}^{t_{j+1}}(1+t)^{\frac{d}{2}}\mathbb{E}_{\nu_\omega}[(\mathcal{P}_t F)^2]dt-2\int_{t_j}^{t_{j+1}}(1+t)^{\frac{d+2}{2}}\mathcal{D}^L_{\nu_\omega}(\mathcal{P}_t F)dt\Big\}. \qquad (3.10)
\end{aligned}$$

In the last line, $\mathcal{D}^L_{\nu_\omega}(\cdot)$ refers to the Dirichlet form (1.11) of associated to the full generator $\mathcal{L}^{\varepsilon}_{\Gamma_L}$. Let $\{\tau_x, x\in\mathbb{Z}^d\}$ denote the group of translation operators on the space of spin fields $\mathbb{R}^{\Gamma_L}$. More precisely, given any $\varphi\in\mathbb{R}^{\Gamma_L}$, set $(\tau_x\varphi)_y=\varphi_{x+y}$. The translations also extend to functions and semigroups by $(\tau_x F)(\varphi)=F(\tau_x\varphi)$ and $(\tau_x\mathcal{P}_t^{\varepsilon,L})(F)=\mathcal{P}_t^{\varepsilon,L}(\tau_x F)$. Since the dynamics is translation invariant and the Gibbs measure ν_ω is product, τ_x commute with $\mathcal{L}^{\varepsilon}_{\Gamma_L}$ and $\mathcal{P}_t^{\varepsilon,L}$. This applies to (3.10) and get

$$\begin{aligned}
&\sum_{j=0}^{n(T)}\Big\{\frac{d+2}{2}\int_{t_j}^{t_{j+1}}(1+t)^{\frac{d}{2}}\mathbb{E}_{\nu_\omega}[(\mathcal{P}_t F)^2]dt-2\int_{t_j}^{t_{j+1}}(1+t)^{\frac{d+2}{2}}\mathcal{D}^L_{\nu_\omega}(\mathcal{P}_t F)dt\Big\}\\
&\quad=\sum_{j=0}^{n(T)}\frac{1}{|\Gamma_L|}\sum_{x\in\Gamma_L}\Big\{\frac{d+2}{2}\int_{t_j}^{t_{j+1}}(1+t)^{\frac{d}{2}}\mathbb{E}_{\nu_\omega}[(\mathcal{P}_t\tau_x F)^2]dt-2\int_{t_j}^{t_{j+1}}(1+t)^{\frac{d+2}{2}}\mathcal{D}^L_{\nu_\omega}(\mathcal{P}_t F)dt\Big\}. \qquad (3.11)
\end{aligned}$$

Notice that translation also preserves nearest-neighbor exchanges of particles, and we have replaced $\mathcal{D}^L_{\nu_\omega}(\mathcal{P}_t F)$ with $\mathcal{D}^L_{\nu_\omega}(\mathcal{P}_t\tau_x F)$ in (3.11). Using the spectral gap inequality for canonical measures (which follows from Theorem 2.1.1), we define a decoupling that is algebraic in time and balances the polynomial LSI constant. For any $t\in[1,T]$, let

$$\ell(t)=\left(\frac{t_{\{\max j;t_j\leq t\}}}{(d+2)R_0}\right)^{\frac{1}{4}}, \qquad (3.12)$$

where R_0 is the constant in (2.1). We define $\mathcal{R}=\{\Omega_1,...,\Omega_q\}$ as the family of disjoint cubes with side length $\lfloor\ell\rfloor$ such that each Ω_j, $1\leq j\leq q$ is a translation of Γ_ℓ and $\cup_{j=1}^q\Omega_j=\Gamma_L$. It is easy to see that $q=O[(L/\ell)^d]$. Moreover, let $\mathbf{M}=(M_1,...,M_q)$ denote a vector in $\mathbb{R}^q$, we define $B_{\ell,L}$ to be the conditional expectation operator on $L^2(\nu_\omega^{\Gamma_L})$ such that

$$B_{\ell,L}F=\mathbb{E}_{\nu_\omega^{\Gamma_L}}[F|M_1,...,M_q],\quad \forall F\in L^2(\nu_\omega^{\Gamma_L}). \qquad (3.13)$$

The idea for (3.13) is that we wish to restrict the spin fluctuations to the hypersurface specified by the multi-canonical product measure $\nu_{M_1}^{\Omega_1}\otimes\cdots\otimes\nu_{M_q}^{\Omega_q}$. Since the Kawasaki dynamics is conservative, the individual diffusions in Ω_j with total spins $M_j=\sum_{x\in\Omega_j}\varphi_x$ are independent of each other. Thus after conditioning, we may apply functional inequalities to each $\nu_{M_j}^{\Omega_j}$. Since the side length $\ell(t)$ is of order $O(t^{1/4})\ll O(\sqrt{t})$, we

may resemble an extra $1+t$ term for the first integrand in (3.11):

$$\begin{aligned}
&\int_{t_j}^{t_{j+1}} (1+t)^{\frac{d}{2}} \mathbb{E}_{\nu_\omega}[(\mathcal{P}_t\tau_x F)^2]dt \\
&\leq 2\Big\{ \int_{t_j}^{t_{j+1}} (1+t)^{\frac{d}{2}} \mathbb{E}_{\nu_\omega}\big(\mathbb{E}_{\nu_{M_1}^{\Omega_1}\otimes\cdots\otimes\nu_{M_q}^{\Omega_q}}[(\mathcal{P}_t\tau_x F - B_{\ell,M}\mathcal{P}_t\tau_x F)^2]\big)dt + \int_{t_j}^{t_{j+1}} (1+t)^{\frac{d}{2}} \mathbb{E}_{\nu_\omega}[(B_{\ell,M}\mathcal{P}_t\tau_x F)^2]dt \Big\} \\
&\leq 2\Big\{ \int_{t_j}^{t_{j+1}} R_0\ell(t)^2(1+t)^{\frac{d}{2}} \sup_{M\in\mathbb{R}^q} \sum_{j=1}^{q} \mathcal{D}_{\nu_{\Omega_j}^{M_j}}(\mathcal{P}_t F)dt + \int_{t_j}^{t_{j+1}} (1+t)^{\frac{d}{2}} \mathbb{E}_{\nu_\omega}[(B_{\ell,M}\mathcal{P}_t\tau_x F)^2]dt \Big\} \\
&\leq \frac{2}{d+2}\int_{t_j}^{t_{j+1}} (1+t)^{\frac{d+2}{2}} \mathcal{D}_{\nu_\omega}^{L}(\mathcal{P}_t F)dt + 2\int_{t_j}^{t_{j+1}} (1+t)^{\frac{d}{2}} \mathbb{E}_{\nu_\omega}[(B_{\ell,M}\mathcal{P}_t\tau_x F)^2]dt.
\end{aligned}$$

The second inequality in the above is an application of the Efron-Stein inequality. Following this computations, as well as (3.10) and (3.11), the Dirichlet term is now canceled and it is sufficient to only consider

$$\sum_{j=0}^{n(T)} \frac{1}{|\Gamma_L|} \sum_{x\in\Gamma_L} \int_{t_j}^{t_{j+1}} (1+t)^{\frac{d}{2}} \mathbb{E}_{\nu_\omega}[(B_{\ell,M}\mathcal{P}_t\tau_x F)^2]dt \tag{3.14}$$

in the following steps of the proof.

Step 2. Multiscale second-order moment at equilibrium.

For fixed $\mathbf{M}\in\mathbb{R}^q$, we denote $\nu_{L,\mathbf{M}} = \nu_{M_1}^{\Omega_1}\otimes\cdots\otimes\nu_{M_q}^{\Omega_q}$ for brevity. At any time $t>0$, let $f_t^{\varepsilon,\mathbf{M}}$ be the Radon-Nikodym derivative:

$$f_t^{\varepsilon,\mathbf{M}} := \frac{d\nu_{L,\mathbf{M}}\mathcal{P}_t^{\varepsilon,L}}{d\nu_\omega^L}, \tag{3.15}$$

where $d\nu_{L,\mathbf{M}}\mathcal{P}_t^{\varepsilon,L}$ is the distribution of the perturbative dynamics at time $t>0$, starting from the equilibrium multicanonical distribution $\nu_{L,\mathbf{M}}$. Due to translation invariance and reversibility, we have

$$B_{\ell,\mathbf{M}}\mathcal{P}_t\tau_x F = \mathbb{E}_{\nu_\omega}(F\cdot\tau_{-x}f_t^{\varepsilon,\mathbf{M}}). \tag{3.16}$$

Here, we emphasize that the expectation on the right hand side does not average over $M\in\mathbb{R}^q$. Using Cauchy-Schwarz inequality, $M\in\mathbb{R}^q$ is averaged with respect to ν_ω in the outside:

$$\mathbb{E}_{\nu_\omega}[(B_{\ell,\mathbf{M}}\mathcal{P}_t\tau_x F)^2] \leq 2\mathbb{E}_{\nu_\omega}\big(\mathbb{E}_{\nu_\omega}[(B_{\ell,\mathbf{M}}F)\cdot(\tau_{-x}f_t^{\varepsilon,\mathbf{M}})]^2\big) + 2\mathbb{E}_{\nu_\omega}\big(\mathbb{E}_{\nu_\omega}[(F-B_{\ell,\mathbf{M}}F)\cdot(\tau_{-x}f_t^{\varepsilon,\mathbf{M}})]^2\big). \tag{3.17}$$

The conditional expectation $B_{\ell,\mathbf{M}}$ and the semigroup $\mathcal{P}_t^{\varepsilon,L}$ are both contractions in $L^2(\nu_\omega)$. This yields

$$\mathbb{E}_{\nu_\omega}\big(\mathbb{E}_{\nu_\omega}[(B_{\ell,\mathbf{M}}F)\cdot(\tau_{-x}f_t^{\varepsilon,\mathbf{M}})]^2\big) = \mathbb{E}_{\nu_\omega}\big([B_{\ell,\mathbf{M}}(\mathcal{P}_t^{\varepsilon,L}\tau_x B_{\ell,\mathbf{M}}F)]^2\big) \leq \mathbb{E}_{\nu_\omega}\big([B_{\ell,\mathbf{M}}F]^2\big). \tag{3.18}$$

In the first identity, we applied the definition for the Radon-Nikodym derivative (3.15). For the second estimate, we first omit the conditional expectation in front of the semigroup. Then, due to translation invariance and contraction, $\tau_x\mathcal{P}_t^{\varepsilon,L}$ is upper bounded by identity.

The remaining part of this step focuses on the estimate for $\mathbb{E}_{\nu_\omega}([B_{\ell,\mathbf{M}}F]^2)$, which is a second-order moment at thermodynamic equilibrium. For $1\leq j\leq q$, define $F_j = \mathbb{E}_{\nu_\omega}(F|M_1,...,M_j)$. Since we have assumed that F is centered, let $F_0 := \mathbb{E}_{\nu_\omega}(F) = 0$. By successive conditioning, we may write

$$\mathbb{E}_{\nu_\omega}([B_{\ell,\mathbf{M}}F]^2) = \sum_{j=0}^{q-1}\mathbb{E}_{\nu_\omega}[(F_{j+1}-F_j)^2] = \sum_{j=0}^{q-1}\mathbb{E}_{\nu_\omega}\big(\mathbb{E}_{\nu_\omega}[(F_{j+1}-F_j)^2|M_1,...,M_j]\big). \tag{3.19}$$

We now estimate $\mathbb{E}_{\nu_\omega}[(F_{j+1}-F_j)^2|M_1,...,M_j]$ with fixed variables $M_1,...,M_j$. Heuristically speaking, we consider $(F_{j+1}-F_j)^2$ to be a measurable function on $\mathbb{R}^{\Omega_{j+1}}$. Let

$$R_j := \mathbb{E}_{\nu_\omega}[F|M_1,...,M_j,\{\varphi_x, x\in\Omega_{j+1}\}]. \tag{3.20}$$

With this notation, we have

$$\mathbb{E}_{\nu_\omega}[(F_{j+1}-F_j)^2|M_1,...,M_j] = \mathbb{E}_{\nu_\omega^{\Omega_{j+1}}}\big[(\mathbb{E}_{\nu_{M_{j+1}}^{\Omega_{j+1}}}[R_j]-\mathbb{E}_{\nu_\omega^{\Omega_{j+1}}}[R_j])^2\big]. \tag{3.21}$$

Since the spectral gap for the Glauber dynamics under the ∞-temperature limit is of order-1, there exists a finite constant $a_0>0$ such that

$$\mathbb{E}_{\nu_\omega^{\Omega_{j+1}}}\big[(\mathbb{E}_{\nu_{M_{j+1}}^{\Omega_{j+1}}}[R_j]-\mathbb{E}_{\nu_\omega^{\Omega_{j+1}}}[R_j])^2\big] \le a_0\sum_{x\in\Omega_{j+1}}\mathbb{E}_{\nu_\omega^{\Omega_{j+1}}}\Big[\big(\partial_x\mathbb{E}_{\nu_{M_{j+1}}^{\Omega_{j+1}}}[R_j]\big)^2\Big]. \tag{3.22}$$

Simple computations of the partial derivatives yields that

$$\partial_x\mathbb{E}_{\nu_{M_{j+1}}^{\Omega_{j+1}}}[R_j] = \frac{1}{|\Gamma_\ell|}\sum_{x\in\Omega_{j+1}}\mathbb{E}_{\nu_{M_{j+1}}^{\Omega_{j+1}}}[\partial_x R_j] - \frac{1}{|\Gamma_\ell|}\mathrm{Cov}_{\nu_{M_{j+1}}^{\Omega_{j+1}}}\Big[R_j,\sum_{x\in\Omega_{j+1}}V'(\varphi_x)\Big]. \tag{3.23}$$

According to the definition (3.20), $\|\partial_x R_j\|_\infty \le \|\partial_x F\|_\infty$. We may assume the validity of the following lemma and conclude the upper bound for the second-order moment at equilibrium:

$$\mathbb{E}_{\nu_\omega}([B_{\ell,\mathbf{M}}F]^2) \le C_0(\log\ell)^2\ell^{-d}\sum_{j=0}^{q-1}\Big(\sum_{x\in\Omega_{j+1}}\|\partial_x R_j\|_\infty\Big)^2 \le C_0(\log\ell)^2\ell^{-d}\,|||\,F\,|||^2\,. \tag{3.24}$$

Lemma 3.2.1. *Let G be a function defined on a cube $\Gamma_\ell=\{1,...,\ell\}^d$ with finite triple norm. There exists a constant $C_0=C_0(d,V)$ such that*

$$\mathrm{Cov}_{\nu_\ell^M}\Big[G,\sum_{x\in\Gamma_\ell}V'(\varphi_x)\Big] \le C_0\log\ell\sum_{x\in\Gamma_\ell}\|\partial_x G\|_\infty. \tag{3.25}$$

To prove this lemma, we take a family of partitions $\{\mathrm{Par}_1,...,\mathrm{Par}_\gamma\}$ of Γ_ℓ. Each partition $\mathrm{Par}_j, 1\le j\le\gamma$ is required to disintegrate Γ_ℓ into a disjoint union of small cubes with side length $\lfloor m^j\rfloor$, where $m\ge 1$ is fixed such that $m^\gamma=\ell$ for some large integer $\gamma\in\mathbb{N}$. Then the left hand side of (3.25) is partitioned in to $\gamma=O(\log\ell)$ summands, with each one being the comparison between scale m^j and m^{j+1}. The parameter m should be small enough such that for each summand, $\mathrm{Var}_{\nu_\ell^M}[\sum_{x\in\Gamma_{m^j}}V'(\varphi_x)-\sum_{x\in\Gamma_{m^{j+1}}}V'(\varphi_x)]$ is upper bounded by a uniform constant. Within this framework, we have eliminated the contribution of $\sum_{x\in\Gamma_\ell}V'(\varphi_x)$ and accumulated an extra factor $\sum_{j=1}^\gamma 1\sim O(\log\ell)$. For the complete proof of this sketch, we refer to [LY03, Lemma 4.1].

Step 3. Bodineau-Helffer inequality.

For the second term $\mathbb{E}_{\nu_\omega}\big(\mathbb{E}_{\nu_\omega}[(F-B_{\ell,\mathbf{M}}F)\cdot(\tau_{-x}f_t^{\varepsilon,\mathbf{M}})]^2\big)$ in (3.17), we can write it as a conditional covariance:

$$\mathbb{E}_{\nu_\omega}\big(\mathbb{E}_{\nu_\omega}[(F-B_{\ell,\mathbf{M}}F)\cdot(\tau_{-x}f_t^{\varepsilon,\mathbf{M}})]^2\big) \le \sup_{\mathbf{M}\in\mathbb{R}^q}\mathrm{Cov}_{\nu_{\mathbf{M}}^L}(F,\tau_{-x}f_t^{\varepsilon,\mathbf{M}})^2. \tag{3.26}$$

Due to a result of Bodineau and Helffer [BH99], we have the following lemma for continuous spins $\varphi\in\mathbb{R}^{\Gamma_L}$.

Lemma 3.2.2 (Continuous Bodineau-Helffer inequality)**.** *Let $\nu_{L,\mathbf{M}}=\nu_{\Omega_1}^{M_1}\otimes\cdots\otimes\nu_{\Omega_q}^{M_q}$ be the probability measure on the Euclidean space $\mathbb{R}^{\Gamma_L}$ satisfying (LSI) with constant $\gamma\propto\ell^{-2}$. Then for any $F\in L^2(\nu_{L,\mathbf{M}})$ and $G\ge 0$ (which refers to the Radon-Nikodym density in our case), we have*

$$\sum_{x\in\Gamma_L}\mathrm{Cov}_{\nu_{L,\mathbf{M}}}(F,\tau_{-x}G)^2 \le C_0\ell^4\,|||\,F\,|||^2\,\mathcal{D}_{\nu_\omega^L}^L(\sqrt{G}), \tag{3.27}$$

where $C_0 = C_0(d, \omega, V)$, and $\mathcal{D}^L_{\nu^L_\omega}(\cdot)$ refers to the Glauber Dirichlet form (1.11) with respect to the grand canonical measure and the generator $\mathcal{L} = -\Delta$.

We refer to Otto and Reznikoff [OR07, Lemma 5] for a simple proof of this result which is similar to the one for Talagrand's inequality. For the critical Ising model, we need a similar estimate with F replaced by the single spin variable σ_0, see Lemma 3.4.1 and the proof therein.

Lemma 3.2.2, together with (3.17), (3.18), (3.24), and (3.26), implies that

$$\begin{aligned}
&\sum_{j=0}^{n(T)} \frac{1}{|\Gamma_L|} \sum_{x\in\Gamma_L} \int_{t_j}^{t_{j+1}} (1+t)^{\frac{d}{2}} \mathbb{E}_{\nu_\omega}[(B_{\ell,M}\mathcal{P}_t\tau_x F)^2]dt \\
&\quad\lesssim \sum_{j=0}^{n(T)} \Big\{ (\log t_j)(t_{j+1}-t_j) \,|||\, F \,|||^2 + t_j \,|||\, F \,|||^2 \, L^{-d} \int_{t_j}^{t_{j+1}} (1+t)^{\frac{d}{2}} \mathcal{D}^\varepsilon_{\nu^L_\omega}(\sqrt{\tau_{-x} f_t^{\varepsilon,\mathbf{M}}}) \Big\}. \qquad (3.28)
\end{aligned}$$

Notice that we have replaced the Glauber-Dirichlet form in (3.27) by the full Dirichlet form for the perturbative generator $\mathcal{L}^\varepsilon_{\Gamma_L}$.

Step 4. Entropy estimate.

Since the density $f_t^{\varepsilon,\mathbf{M}}$ implicitly contains the semigroup $\mathcal{P}_t^{\varepsilon,L}$, we apply the de Bruijn inequality (1.23) to show that

$$\begin{aligned}
t_j \,|||\, F \,|||^2 \, L^{-d} \int_{t_j}^{t_{j+1}} (1+t)^{\frac{d}{2}} \mathcal{D}^\varepsilon_{\nu^L_\omega}(\sqrt{\tau_{-x} f_t^{\varepsilon,\mathbf{M}}}) \qquad (3.29)\\
\lesssim t_j \,|||\, F \,|||^2 \left(\frac{\ell}{L}\right)^d \Big(\mathbb{E}_{\nu^L_\omega}\big[\Phi(f_{t_j}^{\varepsilon,\mathbf{M}})\big] - \mathbb{E}_{\nu^L_\omega}\big[\Phi(f_{t_{j+1}}^{\varepsilon,\mathbf{M}})\big] \Big) \qquad (3.30)\\
\lesssim t_j \,|||\, F \,|||^2 \left(\frac{\ell}{L}\right)^d \mathbb{E}_{\nu^L_\omega}\big[\Phi(f_{t_j}^{\varepsilon,\mathbf{M}})\big]. \qquad (3.31)
\end{aligned}$$

Lemma 3.2.3. *For $f_t^{\varepsilon,\mathbf{M}}$ defined as (3.15), there exists a constant $C_0 = C_0(d,\omega,V)$ such that*

$$\mathbb{E}_{\nu^L_\omega}\big[\Phi(f_{t_j}^{\varepsilon,\mathbf{M}})\big] \le C_0 \Big(\frac{L}{\ell}\Big)^d (\log \ell + \log \varepsilon^{-1}) \qquad (3.32)$$

holds for all $t \ge 1$.

Proof. Let Γ_L/Ω denote the quotient space obtained by collapsing each cube $\Omega_j, 1 \le j \le q$ into a vertex. One can easily construct a nearest-neighbor graph structure and regard Γ_L/Ω as a d-dimensional cube with side length $K = L/\ell$. We now consider dynamics with periodic boundary condition and let $\mathbb{T}_K$ be the d-dimensional torus formed by identifying the opposite boundaries of Γ_L/Ω.

It is intuitive to see that, without expectation $\mathbb{E}_{\nu^{\Gamma_L}_\omega}$, the entropy $\Phi(f_{t_j}^{\varepsilon,\mathbf{M}})$ should approximate the solution to the following initial value problem:

$$\partial_t g_t(\mathbf{N}) = g_t(\mathbf{N}) \mathcal{L}^\varepsilon_{\mathbb{T}_K} g_t(\mathbf{N}), \quad g_0(\mathbf{N}) = \delta_{\mathbf{M}}, \quad \mathbf{N} = (\mathbf{N}_1, ..., \mathbf{N}_q) \in \mathbb{R}^{\mathbb{T}_K}. \qquad (3.33)$$

Indeed, since the initial Radon-Nikodym derivative is singular, we have $\Phi(f_0^{\varepsilon,\mathbf{M}}) = \delta_{\mathbf{M}}$. Due to the presence of the Glauber perturbation in the generator $\mathcal{L}^\varepsilon_{\mathbb{T}_K}$, the discrete parabolic PDE (3.33) has infinite speed of propagation. Hence, we naturally define $g_t(\mathbf{N})$ to be the heat (Gaussian) kernel:

$$g_t(\mathbf{N}) = \frac{1}{\sqrt{\det(2\pi t \Sigma^{-1})}} e^{-\frac{1}{2}(\mathbf{N}-\mathbf{M})\Sigma(\mathbf{N}-\mathbf{M})}. \qquad (3.34)$$

Here, $\Sigma^{-1} = \ell^{d-1}(\ell\varepsilon\,\mathrm{Id} - \Delta)$ with Δ being the matrix of Laplacian on $\mathbb{T}_K$. Analogous to (2.7), we decompose

$$\mathbb{E}_{\nu^L_\omega}\big[\Phi(f_{t_j}^{\varepsilon,\mathbf{M}})\big] = \mathbb{E}_{\nu^L_\omega}\big[f_{t_j}^{\varepsilon,\mathbf{M}} \log(f_{t_j}^{\varepsilon,\mathbf{M}}/g_t)\big] + \mathbb{E}_{\nu^L_\omega}\big[f_{t_j}^{\varepsilon,\mathbf{M}} \log g_t\big]. \qquad (3.35)$$

These two terms are much easier to control quantitatively, see [LY03, Section 6]. □

Recall that $\varepsilon = (1+T)^{-2-\frac{d}{2}}$ and $t_{j+1} - t_j > t_j$, we take (3.29) and (3.32) to upper bound the right-hand side of (3.28) by

$$\sum_{j=0}^{n(T)} \Big\{ (\log t_j)(t_{j+1} - t_j) \,|||\, F \,|||^2 + t_j \,|||\, F \,|||^2 \, (\log \ell + \log \varepsilon^{-1}) \Big\}$$

$$\leq C_0 \,|||\, F \,|||^2 \log(2+T) \sum_{j=0}^{n(T)} (t_{j+1} - t_j) \leq C_0 \,|||\, F \,|||^2 \, (1+T) \log(2+T). \tag{3.36}$$

The claim (3.9) is finally concluded by choosing an appropriate constant C_0.

3.3 Correlation properties for critical Ising model in hypercubes

Now, let's turn to the infinite-volume Glauber dynamics for the critical Ising model in $d \geq 5$. Let P_t denote the Markov semigroup associated with the infinite-volume generator (1.19), due to the reversibility of the infinite-volume dynamics, and the translation invariance of $\mu_{\mathbb{Z}^d}$, the temporal correlation is in fact a second order moment:

$$\langle \sigma_0, P_t \sigma_0 \rangle_{\mathbb{Z}^d} = \langle (P_{t/2} \sigma_0)^2 \rangle_{\mathbb{Z}^d}. \tag{3.37}$$

Similar to the Ginzburg-Landau model, the averaging with respect to the infinite-volume measure $\mu_{\mathbb{Z}^d}$ in (3.37) can be approximated using any large finite-volume subgraphs. Instead of using a large cube as in Section 3.1, We consider a d-dimensional discrete torus $\mathbb{T}_L$ with side-length $2L$. It consists of all the vertices in $\mathbb{Z}^d$ modulo the relation that x, y are identical iff $\sum_{i=1}^d |(x_i - y_i) \mod 2L| = 0$. Roughly speaking, $\mathbb{T}_L$ is obtained from gluing the opposite edges of the discrete hypercube Λ_L. The critical Ising measure $\mu_{\mathbb{T}_L} = \langle \cdot \rangle_{\mathbb{T}_L}$ without external field is said to be under periodic boundary condition. The reason why torus $\mathbb{T}_L$ is of interest in our context is that it maintains the translation invariance property, which holds for $\langle \cdot \rangle_{\mathbb{Z}^d}$ but not for $\langle \cdot \rangle_{\Lambda_L}$.

For Step 2. in the next section, we need to analyze the stationary expectation of our local function $F(\sigma) = \sigma_0$ on finite cubes embedded in $\mathbb{T}_L$. Therefore, we require an additional assumption (1-arm) for the 1-spin expectation $\langle \sigma_0 \rangle^+_{\Lambda_L}$ at the center of the cube Λ_L with $+$ boundary condition.

Assumption 3.3.1. *Let $\langle \sigma_0 \rangle^+_{\Lambda_L}$ denote the one-spin expectation on Λ_L with $+$-boundary condition. For some $\delta \in (0, 1]$,*

$$\langle \sigma_0 \rangle^+_{\Lambda_L} \lesssim L^{-\delta}. \tag{1-arm}$$

For the validity of this assumption, $\delta \leq 1$ was shown by Handa, Heydenreich, and Sakai [HHS19] at $d \geq 5$, conditional on the algebraic decay of the infinite volume two-point function:

$$c(d)|x-y|^{2-d} \leq \langle \sigma_x \sigma_y \rangle_{\mathbb{Z}^d} \leq C(d)|x-y|^{2-d}, \tag{3.38}$$

where $|x-y|$ denotes the Euclidean distance of $x, y \in \mathbb{Z}^d$. The upper bound in (3.38) is due to reflection positivity and the infrared bound of the nearest neighbor Ising model, see [FILS78, FFS76, MMS77]. Using lace expansion techniques, the lower bound in (3.38) was first established by Sakai [Sak07, Theorem 1.3] at sufficiently large dimensions $d \gg 4$. In a recent work by Duminil-Copin and Panis [DCP24], they proved the lower bound for any dimension $d \geq 5$ by random current expansion, together with Griffiths' inequality and MMS inequality (see [MMS77]). Based on the solvability for $d = 2$ [WMTB76] and several numerical predictions, $\delta \leq 1$ is also expected to hold for $d = 2, 3$ and 4.

However, $\delta > 0$ remains largely unsolved in any dimension. Through Edward-Sokal coupling, the finite-volume 1-spin magnetization is equivalent to the probability that 0 connects $\partial\Lambda_L$ in the FK-Ising percolation. In other cases, conditional on the algebraic decay of the two-point function $\mathbb{P}_{p_c}(0 \leftrightarrow x) \asymp |x|^{2-d}$, Kozma and Nachmias [KN11] determined the 1-arm exponent for critical Bernoulli percolation on $\mathbb{Z}^d$ in high dimensions. Using a similar method, Hulshof [H15] determined the 1-arm exponent for critical long-range percolation and branching random walk. For $d > 10$ dimensional critical nearest-neighbor percolations, due to the validity

of lace expansions techniques, the result of Kozma-Nachmias holds, see [FvdH]. Similarly, given the recent result of Duminil-Copin and Panis [DCP24], $\delta = 1$ is expected to be the optimal 1-arm exponent for the $d \geq 5$ critical Ising model [HHS19]. It is not yet possible to study lower dimensional FK-Ising percolations, since the non-mean-field nature renders the estimate for the probability of the appearance of large clusters difficult.

Proposition 3.3.2. *Fix any $d \geq 5$ and $L \gg 1$, let $P_t^{\mathbb{T}_L}$ denote the semigroup of the critical Glauber dynamics on the d-dimensional torus $\mathbb{T}_L$. Under the same setting as Theorem 0.0.2, we have the polynomial rate of decay*

$$\langle (P_t^{\mathbb{T}_L}\sigma_0)^2\rangle_{\mathbb{T}_L} \leq \frac{C_0}{t^\alpha}, \qquad \forall t \geq 0. \tag{3.39}$$

Here $C_0 = C_0(\beta_c, \eta, \delta, d, c_M)$, and α is given by the formula (0.3).

Proof of Theorem 0.0.2. Since the infinite-volume measure and the Glauber semigroup can both be approximated by the finite-volume ones, using an argument similar to the proof of Proposition 3.1.1, we have

$$\langle (P_t\sigma_0)^2\rangle_{\mathbb{Z}^d} = \lim_{L\to\infty} \langle (P_t^{\mathbb{T}_L}\sigma_0)^2\rangle_{\mathbb{T}_L}. \tag{3.40}$$

The desired result (0.2) is easily drawn because the constant C_0 in the upper bound (3.39) is independent of the length L. □

In the proof of Proposition 3.3.2, we need the following lemma, which is an easy consequence of (1-arm).

Lemma 3.3.3. *Consider the critical temperature Ising model on $\Lambda_L = [\![-L, L]\!]^d$ with $+$ boundary condition in any fixed dimension $d \geq 2$. Assume the assumption (1-arm) is satisfied, then there exists a universal constant $c_0(\delta, d) > 0$ such that*

$$\frac{1}{|\Lambda_L|}\sum_{x\in\Lambda_L} \big(\langle\sigma_x\rangle^+_{\Lambda_L}\big)^2 \leq c_0 L^{-(2\delta\wedge 1)}. \tag{3.41}$$

Proof. For any $0 \leq i \leq L$, we define λ_i as the union of faces of the cube centered at 0 with side length $2i+1$. Namely, let $\lambda_i := \{x \in \Lambda_L;\ \max\{|x_1|, ..., |x_d|\} = i\}$. By monotonicity in the volume of the magnetization (see e.g., [FV18, Lemma 3.22]), we have

$$\frac{1}{|\Lambda_L|}\sum_{x\in\Lambda_L} \big(\langle\sigma_x\rangle^+_{\Lambda_L}\big)^2 \lesssim \frac{1}{L^d}\sum_{i=0}^{L}\sum_{x\in\lambda_i} \big(\langle\sigma_x\rangle^+_{\Lambda_L}\big)^2 \leq \frac{1}{L^d}\sum_{i=0}^{L}\sum_{x\in\lambda_i} \big(\langle\sigma_x\rangle^+_{x+\Lambda_{L-i}}\big)^2. \tag{3.42}$$

Here, $x + \Lambda_{L-i}$ is obtained by translating the cube Λ_{L-i} by x. According to (1-arm), the total contribution from $i = 0$ and $i = L$ is at most $O(L^{-d-2\delta} + L^{-1})$. Thus, we only need to bound the average (3.42) with a smaller summation from $i = 1$ to $i = L - 1$. Using (1-arm) again and approximation by integral, we can write

$$\begin{aligned}
\frac{1}{L^d}\sum_{i=1}^{L-1}\sum_{x\in\lambda_i} \big(\langle\sigma_x\rangle^+_{x+\Lambda_{L-i}}\big)^2 &\lesssim \frac{1}{L^d}\sum_{i=1}^{L-1}\frac{i^{d-1}}{(L-i)^{2\delta}} \\
&\leq \frac{1}{L^{2\delta}}\Big(\int_0^{1-1/L}\frac{u^{d-1}}{(1-u)^{2\delta}}du + \frac{1}{L}\int_0^{1-1/L}\Big[\frac{(d-1)u^{d-2}}{(1-u)^{2\delta}} + \frac{2\delta u^{d-1}}{(1-u)^{2\delta+1}}\Big]du\Big) \\
&\lesssim \frac{1}{L^{2\delta}}\Big(\int_{1/L}^1\frac{dr}{r^{2\delta}} + \frac{1}{L}\int_{1/L}^1\frac{dr}{r^{2\delta}} + \frac{1}{L}\int_{1/L}^1\frac{dr}{r^{2\delta+1}}\Big).
\end{aligned} \tag{3.43}$$

For $\delta \in (0, 1]$, the highest order of (3.43) is $O(L^{-(2\delta\wedge 1)})$ and the lemma is concluded. □

3.4 Proof of Proposition 3.3.2

We follow a strategy similar to the one used for the proof of claim 3.9. Heuristically speaking, by domain Markov property, we try to decouple the large dynamics on $\mathbb{T}_L$ into a product dynamics on several smaller blocks $\cup_{i=1}^q \Lambda_\ell^i \approx \mathbb{T}_L$. Here $q \sim (L/\ell)^d$ is the number of the small blocks, and the side length $\ell(t)$ is chosen to grow with an appropriate polynomial rate in time such that it balances with the LSI constant γ_{Λ_ℓ}. However, our situation is different from the conservative dynamics (1.3), where fixing the number of particles in each small block produces independence for different blocks and ergodicity of the canonical measure. Instead, due to the nearest-neighbor interaction of the Ising measure, we need to fix the spin configurations on the boundary subgraph $\cup_{i=1}^q \partial\Lambda_\ell^i$ to obtain mutual conditional independence. Finally, the contribution from the boundary spins is controlled by an estimate for the entropy of the conditioned distribution against the full Ising measure. Since the exponent $D > 1/2$ in the mean-field bound is irrelevant for the proof, we write (2.36) in a simpler way:

$$\gamma_{\Lambda_L}^{-1} \lesssim L^\eta, \quad \eta = \eta(D) > 0. \tag{3.44}$$

Step 1. Dynamical decoupling.

Suppose $\alpha = (2\delta \wedge 1)/2\eta$ as defined by (0.3). For the temporal regime $[1, \infty)$, we fix a partition by inductively taking $t_0 = 1$ and $t_n > 2t_{n-1}$, $n \geq 1$. Given a closed interval $[1, T]$ with $T \gg 1$ arbitrarily large but fixed, let $n(T)$ be the largest integer such that $t_{n(T)-1} < T$, and redefine $t_{n(T)} := T$. By backward Kolmogorov's equation, we have

$$\begin{aligned}
&T^{\alpha+1}\langle (P_T^{\mathbb{T}_L}\sigma_0)^2\rangle_{\mathbb{T}_L} - \langle (P_1^{\mathbb{T}_L}\sigma_0)^2\rangle_{\mathbb{T}_L} \\
&= \int_1^T \partial_t \Big(t^{\alpha+1}\langle (P_t^{\mathbb{T}_L}\sigma_0)^2\rangle_{\mathbb{T}_L}\Big) dt \\
&= \sum_{i=0}^{n(T)-1} \int_{t_i}^{t_{i+1}} (\alpha+1)t^\alpha \langle (P_t^{\mathbb{T}_L}\sigma_0)^2\rangle_{\mathbb{T}_L} dt + 2\int_1^T t^{\alpha+1}\langle (P_t^{\mathbb{T}_L}\sigma_0)(\mathrm{L}_{\mathbb{T}_L} P_t^{\mathbb{T}_L}\sigma_0)\rangle_{\mathbb{T}_L} dt \\
&= \sum_{i=0}^{n(T)-1} \int_{t_i}^{t_{i+1}} (\alpha+1)t^\alpha \langle (P_t^{\mathbb{T}_L}\sigma_0)^2\rangle_{\mathbb{T}_L} dt - 2\int_1^T t^{\alpha+1}\mathrm{D}_{\mathbb{T}_L}(P_t^{\mathbb{T}_L}\sigma_0) dt.
\end{aligned} \tag{3.45}$$

For any $x \in \mathbb{T}_L$, let τ_x denote the translation operator on $\Omega_{\mathbb{T}_L}$, i.e., $(\tau_x\sigma)_y = \sigma_{x+y}$. Since the Ising measure $\mu_{\mathbb{T}_L}$ is translation invariant, the first term in the right hand side of (3.45) is equal to

$$\frac{1}{|\mathbb{T}_L|}\sum_{x\in\mathbb{T}_L}\sum_{i=0}^{n(T)-1}\int_{t_i}^{t_{i+1}} (\alpha+1)t^\alpha \langle (P_t^{\mathbb{T}_L}\sigma_x)^2\rangle_{\mathbb{T}_L} dt. \tag{3.46}$$

Now for any $t \in [1, T]$, define

$$\ell(t) = \left(\frac{1}{c_1(\alpha+1)} t_{\max\{i;\, t_i \leq t\}}\right)^{\frac{1}{2\eta}}, \tag{3.47}$$

where $c_1 > 0$ is the constant factor in the inequality (3.44). As we can see, the function $\ell(t)$ is constant in each subinterval $[t_i, t_{i+1})$, and $\ell^{-(2\delta\wedge 1)}t^\alpha \lesssim 1$. From then on, let ℓ be the shorthand for $\ell(t)$, since the time t for the dynamics is implicitly understood. Using $\lfloor \ell \rfloor$ as side-length, we define a grid (evolving through time) on the torus:

$$\partial\Lambda_\ell := \big\{ x \in \mathbb{T}_L;\ \exists 1 \leq j \leq d, k \in \mathbb{N}\cup\{0\},\ \text{s.t.,}\ |x_j| = (k + \frac{1}{2})(\lfloor \ell \rfloor - 1)\big\}. \tag{3.48}$$

Suppose $\Lambda_\ell := \{\Lambda_\ell^1, ..., \Lambda_\ell^q\}$ is a family of hypercubes with side length $\lfloor \ell \rfloor - 2$, such that the centers $\{z^j\}_{j=1}^q$ of all these cubes satisfy $z^j \in (\lfloor \ell \rfloor - 1)\mathbb{Z}^d \cap \mathbb{T}_L$. Then one easily finds that $\partial\Lambda_\ell$ is the union of the boundaries

of these cubes and $\mathbb{T}_L = \partial\Lambda_\ell \cup \Lambda_\ell^1 \cup \cdots \cup \Lambda_\ell^q$. For simplicity of notation, we would like to omit the ℓ subscripts in the following proof.

Let $B_{\partial\Lambda}$ denote the conditional expectation operator:

$$B_{\partial\Lambda}(u)(\tau) = \mathbb{E}_{\mathbb{T}_L}(u|\ \tau|_{\partial\Lambda}), \quad \forall u \in L^2(\mu_{\mathbb{T}_L}),\ \tau \in \Omega_{\mathbb{T}_L}. \tag{3.49}$$

A simple application of the inequality $x^2 \le 2y^2 + 2(x-y)^2$ yields an upper bound for (3.46):

$$\begin{aligned}\frac{2}{|\mathbb{T}_L|}\sum_{x\in\mathbb{T}_L}\sum_{i=0}^{n(T)-1}\int_{t_i}^{t_{i+1}}(\alpha+1)t^\alpha\langle(B_{\partial\Lambda}P_t^{\mathbb{T}_L}\sigma_x)^2\rangle_{\mathbb{T}_L}dt \\ + \frac{2}{|\mathbb{T}_L|}\sum_{x\in\mathbb{T}_L}\int_1^T(\alpha+1)t^\alpha\langle\big[(I-B_{\partial\Lambda})P_t^{\mathbb{T}_L}\sigma_x\big]^2\rangle_{\mathbb{T}_L}dt,\end{aligned} \tag{3.50}$$

where I denotes the identity operator. For a fixed $\tau \in \Omega_{\mathbb{T}_L}$, the conditional expectation $B_{\partial\Lambda}(\cdot)(\tau)$ is equivalent to the expectation under the product probability measure $\mu_\Lambda(\cdot|\tau)$ defined by

$$\mu_\Lambda(\cdot|\tau) = \mu_{\Lambda^1}^\tau \otimes \cdots \otimes \mu_{\Lambda^q}^\tau.$$

Indeed, this is provided by the domain Markov property of spin systems, and the fact that Ising measure has nearest-neighbor interactions. After fixing the spin configuration on the boundary $\partial\Lambda$, the spins in different cubes $\Lambda^i \ne \Lambda^j$ are independent. Meanwhile, we observe that for any fixed $x \in \mathbb{T}_L$, the variable inside the angle bracket in the second term of (3.50) is in fact the variance of $P_t^{\mathbb{T}_L}\sigma_x$ under the measure $\mu_\Lambda(\cdot|\tau)$. According to the aforementioned reasons and the Efron-Stein inequality, we have

$$\langle[(I-B_{\partial\Lambda})P_t^{\mathbb{T}_L}\sigma_x]^2\rangle_{\mathbb{T}_L} \le \sum_{j=1}^q\langle\mathrm{Var}_{\Lambda^j}^\tau(P_t^{\mathbb{T}_L}\sigma_x)\rangle_{\mathbb{T}_L}. \tag{3.51}$$

Next, we apply (SGI) and (3.44) to all cubes Λ^j, the sum of variances is bounded by

$$\begin{aligned}\mathrm{gap}(\mathrm{L}_{\Lambda^1}^\tau)^{-1}\sum_{j=1}^q\sum_{y\in\Lambda^j}\langle(\nabla_y P_t^{\mathbb{T}_L}\sigma_x)^2\rangle_{\mathbb{T}_L} &\le \mathrm{gap}(\mathrm{L}_{\Lambda^1}^\tau)^{-1}\sum_{y\in\mathbb{T}_L}\langle(\nabla_y P_t^{\mathbb{T}_L}\sigma_0)^2\rangle_{\mathbb{T}_L} \\ &\le c_1\lfloor\ell\rfloor^\eta \mathrm{D}_{\mathbb{T}_L}(P_t^{\mathbb{T}_L}\sigma_0).\end{aligned} \tag{3.52}$$

For the first inequality, we applied the translation invariance of the dynamics again and dropped the dependence on the starting position $x \in \mathbb{T}_L$. The second inequality follows from assumption (3.44), (SGI), and the fact that LSI implies SGI with the same constant. Due to the boundedness of the flipping rates (1.17), the standard Dirichlet form $\sum_{x\in\mathbb{T}_L}\langle(\nabla_x P_t^{\mathbb{T}_L}\sigma_0)^2\rangle_{\mathbb{T}_L}$ is equivalent to the Dirichlet form (1.20) of our dynamics, hence we get (3.52). Combine (3.51), (3.52), and $\lfloor\ell\rfloor^\eta \le \ell^{2\eta}$, the integrand in the second term of (3.50) is upper bounded by $2t^{\alpha+1}\mathrm{D}_{\mathbb{T}_L}(P_t^{\mathbb{T}_L}\sigma_0)$. Thus in order to bound the two integrals in (3.45), it suffices to consider the first term in (3.50).

Step 2. Averaged 1-spin expectation at equilibrium.

We then study the second order moment of the conditioned dynamics $\langle(B_{\partial\Lambda}P_t^{\mathbb{T}_L}\sigma_x)^2\rangle_{\mathbb{T}_L}$. First, apply $x^2 \le 2y^2 + 2(x-y)^2$ again, we get

$$\langle(B_{\partial\Lambda}P_t^{\mathbb{T}_L}\sigma_x)^2\rangle_{\mathbb{T}_L} \le 2\langle\big[B_{\partial\Lambda}P_t^{\mathbb{T}_L}(B_{\partial\Lambda}\sigma_x)\big]^2\rangle_{\mathbb{T}_L} + 2\langle\big[B_{\partial\Lambda}P_t^{\mathbb{T}_L}(\sigma_x - B_{\partial\Lambda}\sigma_x)\big]^2\rangle_{\mathbb{T}_L}. \tag{3.53}$$

Since both the conditioning and the semigroup operators are L^2 contractions, the first term in (3.53) is bounded by $2\langle(B_{\partial\Lambda}\sigma_x)^2\rangle_{\mathbb{T}_L}$. For the second term, we fix a given boundary condition τ and time $t \in [1,T]$, define f_t^τ to be the Radon-Nikodym density function:

$$f_t^\tau := \frac{d\mu_\Lambda(\cdot|\tau)P_t^{\mathbb{T}_L}}{d\mu_{\mathbb{T}_L}}, \tag{3.54}$$

where $d\mu_\Lambda(\cdot|\tau)P_t^{\mathbb{T}_L}$ denotes the law of the Glauber dynamics at time t, started from initial distribution $d\mu_\Lambda(\cdot|\tau)$. This definition leads to

$$\langle \big[B_{\partial\Lambda} P_t^{\mathbb{T}_L}(\sigma_x - B_{\partial\Lambda}\sigma_x)\big]^2\rangle_{\mathbb{T}_L} = \langle \big(\mathbb{E}_{\mathbb{T}_L}\big[f_t^\tau \cdot (\sigma_x - B_{\partial\Lambda}\sigma_x)\big]\big)^2\rangle_{\mathbb{T}_L}. \tag{3.55}$$

To sum up, the first term in (3.50) is bounded by

$$\begin{aligned}\frac{4}{|\mathbb{T}_L|}\sum_{x\in\mathbb{T}_L}\int_1^T (\alpha+1)t^\alpha \langle (B_{\partial\Lambda}\sigma_x)^2\rangle_{\mathbb{T}_L} dt \\ + \frac{4}{|\mathbb{T}_L|}\sum_{x\in\mathbb{T}_L}\sum_{i=0}^{n(T)-1}\int_{t_i}^{t_{i+1}} (\alpha+1)t^\alpha \langle \big(\mathbb{E}_{\mathbb{T}_L}\big[f_t^\tau\cdot(\sigma_x - B_{\partial\Lambda}\sigma_x)\big]\big)^2\rangle_{\mathbb{T}_L} dt.\end{aligned} \tag{3.56}$$

For any fixed τ and $x\in\Lambda^1$, we have $(B_{\partial\Lambda}\sigma_x)(\tau) = \mathbb{E}^\tau_{\mathbb{T}_L\setminus\partial\Lambda}(\sigma_x) = \mathbb{E}^\tau_{\Lambda^1}(\sigma_x)$. This is because the spin function σ_x is only supported in the first cube Λ^1. By the FKG inequality and the symmetry of the torus, we obtain

$$-\mathbb{E}^+_{\Lambda^1}(\sigma_x) = \mathbb{E}^-_{\Lambda^1}(\sigma_x) \le \mathbb{E}^\tau_{\Lambda^1}(\sigma_x) \le \mathbb{E}^+_{\Lambda^1}(\sigma_x). \tag{3.57}$$

Moreover, the cubes in the family Λ are essentially equivalent, thus we get

$$\begin{aligned}\frac{1}{|\mathbb{T}_L|}\sum_{x\in\mathbb{T}_L}\langle (B_{\partial\Lambda}\sigma_x)^2\rangle_{\mathbb{T}_L} &\le \frac{1}{L^d}\Big(\sum_{x\in\Lambda^1} q\big[\mathbb{E}^+_{\Lambda^1}(\sigma_x)\big]^2 + |\partial\Lambda|\Big) \\ &\lesssim \frac{1}{\ell^d}\sum_{x\in\Lambda^1}\big[\mathbb{E}^+_{\Lambda^1}(\sigma_x)\big]^2 + \frac{1}{\ell}.\end{aligned} \tag{3.58}$$

According to Lemma 3.3.3, the first term in (3.58) is of order $O(\ell^{-(2\delta\wedge 1)})$. By the definition of ℓ, there exists a constant $c_2>0$ such that $\ell^{-(2\delta\wedge1)}t^\alpha \le c_2$, and the first integral in (3.56) is therefore bounded by $4c_2(\alpha+1)(T-1)$.

Step 3. Discrete Bodineau-Helffer inequality.

Let $\mathrm{Cov}^\omega_{\mathbb{T}_L\setminus\partial\Lambda}$ be the covariance operator for the measure $\mu_{\mathbb{T}_L}(\cdot|\omega)$. By Jensen's inequality, we have

$$\begin{aligned}\langle \big(\mathbb{E}_{\mathbb{T}_L}\big[f_t^\tau\cdot(\sigma_x - B_{\partial\Lambda}\sigma_x)\big]\big)^2\rangle_{\mathbb{T}_L} &\le \langle \big(B_{\partial\Lambda}\big[f_t^\tau\cdot(\sigma_x - B_{\partial\Lambda}\sigma_x)\big]\big)^2\rangle_{\mathbb{T}_L} \\ &= \langle \big[\,\mathrm{Cov}^\tau_{\mathbb{T}_L\setminus\partial\Lambda}(\sigma_x, f_t^\omega)\big]^2\rangle_{\mathbb{T}_L}.\end{aligned} \tag{3.59}$$

To estimate the average of the covariance terms, we need the following Bodineau-Helffer inequality for the Ising model.

Lemma 3.4.1 (Discrete Bodineau-Helffer inequality)**.** *Consider the Ising model (1.13) on a finite subgraph $\Lambda\subset\mathbb{Z}^d$ with any boundary condition τ. Let $F:\Omega_\Lambda\mapsto\mathbb{R}_+$ be an arbitrary square-integrable non-negative function, we have*

$$\sum_{x\in\Lambda}\mathrm{Cov}^\tau_\Lambda(F,\sigma_x)^2 \le \frac{64c_M}{\gamma_\Lambda^2} D^\tau_\Lambda(\sqrt{F}). \tag{3.60}$$

In particular, c_M denotes the L^∞ upper bound for the flipping rates (see (1.17)). $\mathrm{Cov}^\tau_\Lambda(\cdot,\cdot)$ and $D^\tau_\Lambda(\cdot)$ denote the covariance and standard Glauber Dirichlet form with respect to the Ising measure on Λ. γ_Λ refers to the log-Sobolev constant of the Ising model on Λ.

Proof of Lemma 3.4.1. Since the proof does not involve the geometry of the graph and the boundary condition, we will omit the G subscripts and the τ superscripts for brevity in the following contexts. We also

assume $\int F(\sigma)d\mu = 1$, where μ denotes the Ising measure on Λ. Since μ is the unique invariant measure of the dynamics, we know that $P_\infty F = P_0 F = \int F(\sigma)d\mu = 1$. By straightforward calculation, we have

$$\begin{aligned}\mathrm{Cov}(F,\sigma_x) &= \int \sigma_x\big[(P_0F)(\sigma) - (P_\infty F)(\sigma)\big]d\mu \\ &= -\int\int_0^\infty \sigma_x(\mathrm{L}P_tF)(\sigma)d\mu dt \\ &= -\int\int_0^\infty \sigma_x\, c(x,\sigma)\big[(P_tF)(\sigma^x) - (P_tF)(\sigma)\big]d\mu dt. \end{aligned} \tag{3.61}$$

For the last equality, we applied the self-adjointness of the Glauber generator L and canceled the sum over other vertices. Now we use the identity $a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$ for positive constants a, b and the detailed balance condition for the flipping rates to get

$$\begin{aligned}|\,\mathrm{Cov}(F,\sigma_x)| &\le \int_0^\infty\int c(x,\sigma)|\nabla_x P_tF(\sigma)|d\mu dt \\ &= \int_0^\infty\int c(x,\sigma)|\nabla_x\sqrt{P_tF}(\sigma)| \times |\sqrt{P_tF(\sigma^x)} + \sqrt{P_tF(\sigma)}|d\mu dt \\ &\le 2c_M^{1/2}\int_0^\infty\int c(x,\sigma)^{1/2}|\nabla_x\sqrt{P_tF}(\sigma)| \times \sqrt{P_tF(\sigma)}d\mu dt. \end{aligned} \tag{3.62}$$

For any $\varepsilon \in (0,\gamma)$, we take the square and sum over $x \in G$ to get

$$\begin{aligned}\sum_{x\in\Lambda}\mathrm{Cov}(F,\sigma_x)^2 &\le 4c_M\sum_{x\in\Lambda}\Big(\int_0^\infty\Big[\int c(x,\sigma)|\nabla_x\sqrt{P_tF}(\sigma)|^2d\mu\Big]^{1/2}\Big[\int (P_tF)(\sigma)d\mu\Big]^{1/2}dt\Big)^2 \\ &= 4c_M\sum_{x\in\Lambda}\frac{1}{\varepsilon^2}\Big(\int_0^\infty \varepsilon e^{\varepsilon t}\Big[\int e^{-2\varepsilon t}c(x,\sigma)|\nabla_x\sqrt{P_tF}(\sigma)|^2d\mu\Big]^{1/2}dt\Big)^2 \\ &\le 8c_M\sum_{x\in\Lambda}\frac{1}{\varepsilon^2}\int_0^\infty \mathrm{D}(\sqrt{P_tF})de^{\varepsilon t} \\ &\le -8c_M\sum_{x\in\Lambda}\int_0^\infty \frac{1}{\varepsilon}e^{\varepsilon t}\frac{d}{dt}\mathbb{E}_\mu\Phi(P_tF)dt. \end{aligned} \tag{3.63}$$

In particular, for the derivation of (3.63), we reapplied the de Bruijn identity: $\partial_t\mathbb{E}_\mu\Phi(P_tF) = \int P_tF\mathrm{L}\log P_tF d\mu \le -\mathrm{D}(\sqrt{P_tF})$. Thus, the inequality $\mathbb{E}_\mu\Phi(P_tF) \le e^{-\gamma t}\mathbb{E}_\mu\Phi(F)$ (which follows from (LSI)) and an easy integration by parts in time together imply

$$\begin{aligned}\sum_{x\in\Lambda}\mathrm{Cov}(F,\sigma_x)^2 &\le \frac{8c_M}{\varepsilon}\mathbb{E}_\mu\Phi(F) + 8c_M\int_0^\infty e^{\varepsilon t}e^{-\gamma t}\mathbb{E}_\mu\Phi(F)dt \\ &= 8c_M\Big(\frac{1}{\varepsilon} + \frac{1}{\gamma-\varepsilon}\Big)\mathbb{E}_\mu\Phi(F). \end{aligned} \tag{3.64}$$

Optimizing in $\varepsilon \in (0,\gamma)$ completes the proof of the inequality:

$$\sum_{x\in\Lambda}\mathrm{Cov}(F,\sigma_x)^2 \le \frac{32c_M}{\gamma}\mathbb{E}_\mu\Phi(F) \le \frac{64c_M}{\gamma^2}\mathrm{D}(\sqrt{F}). \tag{3.65}$$

□

Remark 3.4.2. A similar upper bound for the sum of squared covariances should also hold for stochastic dynamics in the continuum scaling limit, provided that the jump rates are bounded uniformly in the scaling parameter.

Step 4. Entropy estimate.

According to Lemma 3.4.1 and (3.59), and the fact that Dirichlet form (or, equivalently, the standard Dirichlet form) is monotonically increasing in the size of the graphs, the second term of (3.56) is upper bounded by

$$\begin{aligned}&\frac{256c_Mc_1^2}{L^d}\sum_{i=0}^{n(T)-1}\int_{t_i}^{t_{i+1}}(\alpha+1)t^\alpha\lfloor\ell\rfloor^{2\eta}\langle \mathrm{D}_{\mathbb{T}_L}(\sqrt{f_t^\tau})\rangle_{\mathbb{T}_L}dt\\&\qquad\le\frac{256(\alpha+1)c_Mc_1^2}{L^d}\sum_{i=0}^{n(T)-1}t_i^\alpha\lfloor\ell(t_i)\rfloor^{2\eta}\langle\mathbb{E}_{\mathbb{T}_L}\Phi(f_{t_i}^\tau)\rangle_{\mathbb{T}_L}.\end{aligned}\tag{3.66}$$

Here, $\mathbb{E}_{\mathbb{T}_L}\Phi(f_{t_i}^\tau):=\mathbb{E}_{\mathbb{T}_L}(f_{t_i}^\tau\log f_{t_i}^\tau)$ denotes the entropy of the non-negative function $f_{t_i}^\tau$. In the above, the second inequality follows from the de Bruijin identity: $\partial_t\mathbb{E}_{\mathbb{T}_L}\Phi(f_t^\tau)=\int f_t^\tau\mathrm{L}\log f_t^\tau d\mu_{\mathbb{T}_L}\le-\mathrm{D}_{\mathbb{T}_L}(\sqrt{f_t^\tau})$. Since the entropy of the Glauber dynamics is decreasing in time, we may trivially bound it by $\mathbb{E}_{\mathbb{T}_L}\Phi(f_0^\tau)$. Therefore, we apply definition (3.54) again and calculate

$$\begin{aligned}\mathbb{E}_{\mathbb{T}_L}\Phi(f_0^\tau)&=\int d\mu_{\mathbb{T}_L}(\cdot|\tau)\log\frac{d\mu_{\mathbb{T}_L}(\cdot|\tau)}{d\mu_{\mathbb{T}_L}}\\&=\int d\mu_{\mathbb{T}_L}(\cdot|\tau)\Big(\sum_{\substack{x\sim y\\x\in\partial\Lambda,\ y\notin\partial\Lambda}}(\sigma_y\tau_x-\sigma_y\sigma_x)+\log\frac{Z}{Z^\tau}\Big),\end{aligned}\tag{3.67}$$

where Z and Z^τ are the partition functions for $\mu_{\mathbb{T}_L}$ and $\mu_{\mathbb{T}_L}(\cdot|\tau)$ respectively. For extra splittings, we define Z_j^τ to be the partition function for the partially conditioned measure $\mu_{\mathbb{T}_L}(\cdot|\tau|_{\partial\Lambda^1\cup\cdots\cup\partial\Lambda^j})$, $1\le j\le q$. (3.67) can be further bounded by

$$\sup_{\tau\in\Omega_{\mathbb{T}_L}}\sum_{j=0}^{q-1}\log\frac{Z_j^\tau}{Z_{j+1}^\tau}\le\sum_{j=0}^{q-1}\log\Big(\sum_{\sigma_x,\ x\in\partial\Lambda^{j+1}}e^{2d\beta_c}\Big)\le c_3\sum_{j=0}^{q-1}\ell^{d-1}\le c_4\ell^{-1}L^d.\tag{3.68}$$

Thus, we get an entropy estimate $\langle\mathbb{E}_{\mathbb{T}_L}\Phi(f_{t_i}^\tau)\rangle_{\mathbb{T}_L}\le c_4\ell(t_i)^{-1}L^d$ for all $i=1,...,n(T)-1$, and henceforth an upper bound for (3.66):

$$256c_Mc_1c_4\sum_{i=0}^{n(T)-1}t_i^{\alpha+1}\ell(t_i)^{-1}\le256c_Mc_1c_5(T-1).\tag{3.69}$$

This is due to $t_{i+1}-t_i>t_i$, as well as the fact that we can find a constant c_5 such that $c_4t_i^\alpha\ell(t_i)^{-1}\le c_5$. Finally, we insert the entropy bound (3.69) and the averaged 1-spin expectation bound (3.58) into (3.56). The proof is completed by choosing $C_0=\max\{4c_2(\alpha+1),256c_Mc_1c_5\}$.

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